

ModularRep

Mathematical Library Manual

Representation theory in Lean

This manual develops the theory of finite group representations, ordinary and Brauer characters, reduction, blocks and weights. It explains the mathematical constructions and their proofs, with references to the corresponding Lean definitions and theorems.

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Using this manual

The chapters follow the constructions needed to pass from representations to local weights. Normal subgroups and quotient groups come first, followed by ordinary characters, Brauer characters and exact Grothendieck groups. These provide the setting for reduction, blocks and weights. The remaining chapters study equivariant bijections, character counts from finite matrices, and several general arguments involving bases, stabilisers and rational series.

Each mathematical statement specifies its coefficient fields and any required assumptions on splitting fields, roots of unity and compatibility. The exposition includes standard results used by the library. Implementation paragraphs distinguish results imported from Mathlib, definitions and deductions in this library, and results supplied as explicit hypotheses to the Lean code. References support the statements at the indicated locations, with version distinctions where these affect the formulation.

`ModularRep.Library` imports a selection of basic constructions. For a particular result, use the module named in its implementation paragraph. The accompanying documentation index lists the source modules and their declarations. Instructions for compiling the library and checking the proofs are supplied with the package. Unless explicitly qualified, theorem and equation numbers refer to this manual.

The general library and the manuscript applications have separate hypotheses. The formalisation report describes the assumptions used by the applications. The Lean code does not construct an instance satisfying all the hypotheses of the manuscript applications.

Chapter 1

Conventions and constructions

The manual treats ordinary characters, Brauer characters, reduction and blocks, and the weights defined from ordinary characters of normaliser quotients.

Three kinds of construction recur in these subjects: passage to a normal subgroup or quotient, passage from representations to their characters or Grothendieck classes, and transport under a group action. The chapters develop these constructions together with the compatibility statements needed to combine them.

Throughout the manual, groups are finite unless a statement says otherwise. The modular prime is ℓ , k has characteristic ℓ , and K has characteristic zero. A discrete valuation ring is a local principal ideal domain that is not a field. Coefficient rings, splitting conditions, algebraic closure and choices of roots are specified where needed. The source files often use the letter p for the modular prime. In the Conlon argument, p denotes a potentially different prime used for permutation lattices.

1.1 From a result to its source

Mathlib supplies the representation types `Representation` and `FDRRep`, the ordinary trace character `Representation.character`, and the underlying group theory, linear algebra and category theory. `ModularRep` defines the finite root correspondences, Brauer characters and exact Grothendieck group used here. The implementation paragraphs explain which steps use Mathlib results and identify the definitions and proofs in the modules of this package.

The accompanying module index, linked from the package reading guide, lists the library and formalisation modules, their declarations and stated assumptions. Its entries are classified by role and status. Examples in this manual illustrate the stated mathematical results.

For local quotient arguments, one can begin with

```
import ModularRep.PCore
import ModularRep.NormalCoreTransport
```

The normaliser quotient results apply to a surjective homomorphism whose kernel lies in the relevant subgroup. The induced isomorphism of normaliser quotients determines the pullback of representations. For reduction, the imports

```
import ModularRep.BrauerDecompositionMap
import ModularRep.KZeroTwist
```

provide the decomposition homomorphism for the chosen modular system and root correspondence. Its hypotheses relate reduction of a stable lattice to Brauer characters. The naturality theorems use the same choices. The modules

```
import Formalisation.FibreTransport
import Formalisation.IndexedAssembly
```

construct equivariant bijections of finite sets from equivariant bijections of their fibres. The chapters state the hypotheses of each construction.

1.2 Hypotheses and proofs

The mathematical exposition includes standard results used by the library. An implementation paragraph says whether the corresponding result is proved in Lean or supplied as an input. In particular, Hypotheses 7.8 and 7.17 collect standard published facts used as inputs to the Lean arguments. Other hypotheses express compatibility of chosen data, as in Hypothesis 6.5. For example, multiplicativity of the central Brauer map follows from cancellation of nontrivial ℓ -group orbits. The construction of the decomposition homomorphism uses an explicit hypothesis relating stable reduction to Brauer characters. The existence of a stable lattice and the deductions from that compatibility hypothesis are proved separately.

An equivariant bijection of finite sets does not by itself preserve a specified relation. To construct a bijection preserving a relation from bijections of the fibres, the latter must preserve the relation and the relation must be invariant under the acting group. Character and block relations require their corresponding compatibility hypotheses.

Finite computations also require a mathematical interpretation. A matrix rank calculation, for example, counts irreducible Brauer characters only after its rows have been identified with character restrictions and the required spanning and independence statements have been established. Chapter 10 states these hypotheses and proves the resulting rank formulas.

Chapter 2

Normal subgroups and local quotients

Throughout this chapter, G and H are finite groups and ℓ is a prime. For a subgroup $Q \leq G$, write $N_G(Q)$ for its normaliser. Since Q is normal in $N_G(Q)$, the quotient $N_G(Q)/Q$ is defined. We study these quotients and radical subgroups under isomorphisms and surjective homomorphisms. They will occur in the definition of a weight in Chapter 8.

2.1 The largest normal ℓ -subgroup

Definition 2.1. The ℓ -core $O_\ell(G)$ is the largest normal ℓ -subgroup of G . A subgroup $Q \leq G$ is ℓ -radical if $Q = O_\ell(N_G(Q))$.

To see that the core exists, take the subgroup generated by all normal ℓ -subgroups of G . Indeed, the product of two normal ℓ -subgroups is normal and has order a power of ℓ . Finiteness reduces the generated subgroup to such a product. It contains every normal ℓ -subgroup and is characterised by this property. An automorphism permutes the normal ℓ -subgroups, so it preserves the core. More generally, a group isomorphism $\phi : G \rightarrow H$ satisfies $\phi(O_\ell(G)) = O_\ell(H)$.

In particular, a radical subgroup is an ℓ -group. If $\phi : G \rightarrow H$ is an isomorphism, then $\phi(N_G(Q)) = N_H(\phi(Q))$, so the equality of cores shows that Q is radical if and only if $\phi(Q)$ is radical.

Implementation.

`ModularRep.PCore` defines the core using Mathlib's subgroup lattice and its theorem on suprema of normal ℓ -subgroups. The basic core results include `ModularRep.pCore_isPGroup`, `ModularRep.pCore_normal` and `ModularRep.normal_pSubgroup_le_pCore`. `ModularRep.RadicalSubgroup` defines radicality by mapping the core of $N_G(Q)$ into G along the normaliser inclusion. `ModularRep.RadicalTransport` proves invariance under isomorphisms.

Proposition 2.2 (The core lies in every radical subgroup). *Every ℓ -radical subgroup of G contains $O_\ell(G)$.*

Proof. Let $P = O_\ell(G)$ and let Q be radical. Both are ℓ -groups, and normality of P makes PQ an ℓ -group. Suppose that $Q < PQ$. The normaliser condition in a finite ℓ -group gives $x \in N_{PQ}(Q) \setminus Q$. Write $x = yz$, with $y \in P$ and $z \in Q$. Both x and z normalise Q , hence $y \in P \cap N_G(Q)$. This intersection is a normal ℓ -subgroup of $N_G(Q)$ and therefore lies in $O_\ell(N_G(Q)) = Q$. Thus $x = yz \in Q$, a contradiction. It follows that $PQ = Q$ and $P \leq Q$. $\square$

Implementation.

The theorem is `ModularRep.pCore_le_of_isRadicalSubgroup` in `ModularRep.NormalCoreTransport`. It uses Mathlib's nilpotence theorem for finite ℓ -groups and its normaliser condition for nilpotent groups.

2.2 Surjections and normaliser quotients

Proposition 2.3 (Normaliser quotients under a surjection). *Let $f : G \twoheadrightarrow H$ be surjective, and let $Q \leq G$ contain $\ker f$. Then $f(N_G(Q)) = N_H(f(Q))$, and the map $gQ \mapsto f(g)f(Q)$ induces an isomorphism*

$$N_G(Q)/Q \simeq N_H(f(Q))/f(Q).$$

Proof. The inclusion $f(N_G(Q)) \leq N_H(f(Q))$ follows by applying f to a conjugate of Q . Conversely, choose $h \in N_H(f(Q))$ and lift it to $g \in G$. Since Q contains the kernel, $Q = f^{-1}(f(Q))$. Conjugation by g therefore preserves Q , proving normaliser surjectivity.

Compose this normaliser map with the quotient by $f(Q)$. An element g of $N_G(Q)$ is in its kernel exactly when $f(g) \in f(Q)$, equivalently when $g \in Q$. The first isomorphism theorem gives the assertion, with the stated formula on cosets. $\square$

Thus containment of $\ker f$ in Q suffices to identify the normaliser quotients. To preserve radicality as well, the following result assumes that $\ker f$ is an ℓ -group.

Proposition 2.4 (Radical subgroups under a surjection). *Suppose that $f : G \twoheadrightarrow H$ has ℓ -group kernel. Then $f(O_\ell(G)) = O_\ell(H)$. For every $Q \leq G$ containing $\ker f$, Q is ℓ -radical if and only if $f(Q)$ is ℓ -radical. Consequently image and inverse image give inverse correspondences between the radical subgroups of G and those of H .*

Proof. The image of the core is a normal ℓ -subgroup, so $f(O_\ell(G)) \leq O_\ell(H)$. The inverse image of $O_\ell(H)$ is an extension of two ℓ -groups and is normal in G . It lies in $O_\ell(G)$, proving equality.

The restriction of f to $N_G(Q)$ is surjective onto $N_H(f(Q))$ by Proposition 2.3, and its kernel is an ℓ -group. Applying the core equality to this restriction identifies the images of the two normaliser cores. Both Q and $O_\ell(N_G(Q))$ contain $\ker f$. Image under f is injective on subgroups containing the kernel, so equality of their images is equivalent to equality of these subgroups. This proves the radical equivalence.

Finally, $\ker f \leq O_\ell(G)$, so every radical subgroup of G contains $\ker f$ by Proposition 2.2. The subgroup correspondence theorem now gives the two inverse maps. $\square$

Implementation.

`ModularRep.NormalCoreTransport` proves the core image equality, normaliser surjectivity and the radical equivalence. Its declarations are `pCore_map_surjective_of_ker_isPGroup`, `normalizerMap_surjective`, `isRadicalSubgroup_iff_map_surjective_of_ker_le` and `normalizerQuotientEquivOfSurjectiveOfKerLE`, all in `ModularRep`. These constructions use Mathlib's subgroup correspondence and quotient homomorphisms.

Example 2.5 (Quotient by the core). Set $P = O_\ell(G)$. The quotient G/P has trivial ℓ -core, and its radical subgroups correspond to Q/P for radical subgroups Q of G . For each such Q , the normaliser quotient is unchanged up to the isomorphism of Proposition 2.3. A representation of this local quotient can therefore be transported without changing its dimension or irreducibility. Chapter 8 explains the resulting construction on weights. If G is itself an ℓ -group, Proposition 2.2 shows that its only radical subgroup is G .

2.3 Central quotients

The preceding quotient results apply to subgroups containing the kernel. A central kernel of order prime to ℓ gives a different subgroup correspondence.

Proposition 2.6 (Quotients by central ℓ' -subgroups). *Let $f : G \rightarrow H$ be a surjective homomorphism of finite groups. Suppose that $Z = \ker f$ is central and $\ell \nmid |Z|$. Then $Q \mapsto f(Q)$ is a bijection between the ℓ -subgroups of G and those of H . For each ℓ -subgroup $R \leq H$, the unique ℓ -subgroup mapping onto R is $O_\ell(f^{-1}(R))$. For every ℓ -subgroup $Q \leq G$,*

$$f^{-1}(N_H(f(Q))) = N_G(Q), \quad Q \text{ is radical} \iff f(Q) \text{ is radical}.$$

Proof. Let $R \leq H$ be an ℓ -subgroup. A Sylow ℓ -subgroup Q of $f^{-1}(R)$ has order $|R|$, since the kernel has order prime to ℓ . Since $Q \cap Z = 1$, its image is R , and $f^{-1}(R) = QZ$. Centrality of Z makes Q normal in this inverse image, so Q lies in its ℓ -core. That core also maps onto R .

The restriction of f to any ℓ -subgroup is injective, since its kernel has order dividing both an ℓ -power and $|Z|$. If two ℓ -subgroups have image R , they both lie in the ℓ -core of $f^{-1}(R)$. Injectivity on that core then shows that they are equal. This proves the correspondence and the formula for its inverse.

An element $g \in G$ normalises Q precisely when $f(g)$ normalises $f(Q)$. For the reverse implication, use uniqueness of the lift on Q and gQg^{-1} . This proves the normaliser formula and surjectivity of $N_G(Q) \rightarrow N_H(f(Q))$.

Uniqueness also implies that the lift of a normal ℓ -subgroup is normal. Consequently $f(O_\ell(G)) = O_\ell(H)$. Apply the same argument to the surjection of normalisers. The image of $O_\ell(N_G(Q))$ is $O_\ell(N_H(f(Q)))$. Injectivity on ℓ -subgroups now proves the radicality equivalence. $\square$

In particular, the induced map on local quotients is

$$N_G(Q)/Q \longrightarrow N_H(f(Q))/f(Q).$$

Its kernel is $QZ/Q \simeq Z$, since $Q \cap Z = 1$. Quotienting by this kernel gives

$$N_G(Q)/(QZ) \cong N_H(f(Q))/f(Q).$$

The two local quotient orders therefore have equal ℓ -parts. Inflation along this map gives characters of $N_G(Q)/Q$ that are trivial on QZ/Q .

Taking f to be the quotient map by a central ℓ' -subgroup Z gives, for every ℓ -radical subgroup Q ,

$$QZ/Z \text{ is radical in } G/Z, \quad N_{G/Z}(QZ/Z) = N_G(Q)/Z.$$

These conclusions also follow from [7, Lemma 2.3(c) and its proof]. Proposition 2.3 does not apply directly to Q , because Z need not be contained in Q .

Implementation.

The subgroup correspondence is proved in `ModularRep.PaperProofs.SporadicFi24P3Definition44NamedCarrierCentralPrimeToPSubgroups`, including `existsUnique_pSubgroup_lift` and `pSubgroupLift_map`. The module whose name ends in `CentralPrimeToRadicals` proves `normalizer_comap_of_central_primeTo` and `radical_iff_map_of_central_primeTo`. The theorem `fixedCentralQuotientSource` gives the quotient specialisation as the structure `ModularRep.NavarroTiep23cFixedCentralQuotientSource`, defined in `ModularRep.PaperProofs.NavarroTiep23cFixedCentralQuotientSource`. `ModularRep.PaperProofs.CentralEllPrimeWeightLocalQuotient` defines the homomorphism `qW` on normaliser quotients and computes its kernel. Its theorems on surjectivity and the ℓ -parts of the quotient orders are stated for radical Q and use the quotient specialisation above. The general subgroup correspondence also allows the trivial subgroup.

2.4 Cyclic quotients and Sylow subgroups

Proposition 2.7 (Cyclic quotients by the core). *If $G/O_\ell(G)$ is cyclic, then $H/O_\ell(H)$ is cyclic for every subgroup $H \leq G$. If G is cyclic, $G/O_\ell(G)$ has order prime to ℓ .*

Proof. Let $I = H \cap O_\ell(G)$. This is a normal ℓ -subgroup of H , so $I \leq O_\ell(H)$. Restriction of the quotient map embeds H/I in $G/O_\ell(G)$. Thus H/I is cyclic, as is its quotient $H/O_\ell(H)$. For the second assertion, the unique Sylow ℓ -subgroup of a finite cyclic group is normal and equals its ℓ -core. Its index is prime to ℓ . $\square$

For a finite group the condition that $G/O_\ell(G)$ be cyclic is equivalent to ℓ -hypoelementarity, that is, the existence of a normal ℓ -subgroup with cyclic quotient of order prime to ℓ . For this terminology, see, for example, [2, Definition 5.5.3]. Indeed, the inverse image of the Sylow ℓ -subgroup of a cyclic quotient would be a normal ℓ -subgroup of G . Maximality of the core forces that quotient Sylow subgroup to be trivial. Thus the cyclic quotient has ℓ' -order. The prime defining hypoelementarity is specified independently of any coefficient field.

Implementation.

`ModularRep.subgroup_quotient_pCore_isCyclic` and
`ModularRep.not_dvd_card_quotient_pCore_of_isCyclic`. The latter belongs to
`ModularRep.CyclicPCore`.

Proposition 2.8 (A criterion for cyclic Sylow subgroups). *Let $T \leq G$ be cyclic. If $|G|_\ell$ divides $|T|$, then G has a cyclic Sylow ℓ -subgroup, and every ℓ -subgroup of G is cyclic. Here $|G|_\ell$ denotes the largest power of ℓ dividing $|G|$.*

Proof. The cyclic group T contains a subgroup of order $|G|_\ell$. It is a Sylow subgroup of G by its order. Every ℓ -subgroup of G is contained in a Sylow subgroup, and all Sylow subgroups are conjugate. Hence each is contained in a cyclic group and is cyclic. $\square$

Implementation.

`ModularRep.CyclicSylow`, in particular
`exists_cyclic_sylow_of_full_part_dvd_card_cyclic_subgroup` and
`isCyclic_of_isPGroup_of_full_part_dvd_card_cyclic_subgroup` in that module's
namespace. They combine Mathlib's Sylow existence, containment and conjugacy theorems with
transport of cyclicity.

Example 2.9 (Applying the order criterion). Suppose G has a cyclic subgroup T and an order factorisation $|G| = ab$ with $\ell \nmid a$ and $b \mid |T|$. Then $|G|_\ell \mid b$, so the preceding proposition gives cyclicity of every ℓ -subgroup of G . This divisibility is proved by `ModularRep.CyclicSylow.orderProj_mul_dvd_right_of_not_dvd_left`.

Chapter 3

Ordinary representations and characters

Pullback, inflation and tensoring with a linear character relate representations of different groups or give new representations of the same group. Their character formulas follow from the representing operators. We first show how characters determine irreducible representations, then study these operations and extension across a cyclic quotient.

Throughout, K is a field of characteristic zero and G is a finite group. A representation ρ of G on a finite dimensional K -vector space V has character $\chi_\rho(g) = \text{tr}_K(\rho(g))$. We state algebraic closure explicitly where it is used. Character separation for finite groups also holds over an arbitrary field of characteristic zero, as the averaging argument below shows. Modular systems and reduction are introduced in Chapter 6.

3.1 Representations and their characters

Definition 3.1. A representation of G on V is a homomorphism $\rho : G \rightarrow \text{GL}_K(V)$. It is irreducible if $V \neq 0$ and its only invariant subspaces are 0 and V . An equivalence $\rho \simeq \sigma$ is a linear isomorphism $T : V \rightarrow W$ satisfying

$$T\rho(g) = \sigma(g)T \quad (g \in G).$$

The set $\text{Irr}_K(G)$ consists of the functions $\chi : G \rightarrow K$ afforded by finite dimensional irreducible K -representations.

Thus equality in $\text{Irr}_K(G)$ is equality at every group element. Choosing different bases or isomorphic realisations does not create new characters: conjugate operators have equal traces. The value at the identity is $\chi_\rho(1) = \dim_K V$, regarded as an element of K . Every character is constant on conjugacy classes, since $\rho(xgx^{-1})$ is conjugate to $\rho(g)$.

Implementation.

Mathlib supplies `Representation`, `Representation.IsIrreducible` and the trace function `Representation.character`, together with the identities `Representation.char_one`, `Representation.char_conj` and `Representation.char_iso`. The additional type `ModularRep.OrdinaryIrreducibleCharacter.Irr` collects the functions admitting an irreducible realisation on some K^n , expressed by `ModularRep.OrdinaryIrreducibleCharacter.Realisation` using these Mathlib definitions. Equality of these character functions is pointwise, as expressed by `ModularRep.OrdinaryIrreducibleCharacter.ext`.

Proposition 3.2 (Characters determine irreducible representations). *Assume that K is algebraically closed. If two finite dimensional irreducible K -representations ρ and σ have equal characters, then $\rho \simeq \sigma$.*

For the corresponding statement about traces of simple modules over a finite dimensional algebra, see, for example, [6, Theorem 1.19].

Proof. Regard the two representation spaces as simple KG -modules. Equality of their traces on G implies equality on every element of KG by linearity. Suppose that the simple modules are nonisomorphic. Schur's lemma says that their mutual homomorphism spaces are zero and their endomorphisms are scalars. The density theorem applied to their direct sum therefore supplies an element of KG acting as an operator of trace one on the first module and as zero on the second. This contradicts equality of traces.

An operator of trace one can be obtained by choosing a basis and taking the projection onto one basis vector. Using this operator also makes the same separation argument valid in positive characteristic, where the trace of the identity may be zero. $\square$

Implementation.

The theorem is

`ModularRep.ModularTraceSeparation.representation_equiv_of_character_eq`. Mathlib supplies the Schur and density statements used in the proof, `IsSimpleModule.algebraMap_end_bijective_of_isAlgClosed` and `Module.Finite.toModuleEnd_moduleEnd_surjective`. The local lemmas `exists_trace_action_ne_of_not_linearEquiv` and `linearMap_trace_surjective`, in the same namespace, make the separating operator explicit. The theorem itself applies to a group over an algebraically closed field of any characteristic and does not require the group to be finite. Its use here is in characteristic zero.

For a finite group, equality of irreducible characters also implies equivalence over an arbitrary field K of characteristic zero. The character averaging formula (see, for example, [8, Section 12.1, proof of Proposition 32]) gives

$$\dim_K \operatorname{Hom}_{KG}(V, W) = \frac{1}{|G|} \sum_{g \in G} \chi_V(g^{-1}) \chi_W(g),$$

where the integer on the left is viewed in K . If $\chi_V = \chi_W$, the right hand side is also $\dim_K \operatorname{End}_{KG}(V)$. This dimension is positive because $V \neq 0$ and its identity map is nonzero. Thus there is a nonzero homomorphism $V \rightarrow W$, which is an isomorphism by irreducibility. The proof does not require that the endomorphism ring consist of scalars.

Implementation.

The theorem `ModularRep.PaperProofs.TypeBOrdinaryTraceSeparationSplitting.nonempty_equiv_of_character_eq` proves this implication using Mathlib's character averaging formula. It applies to every finite group over every field of characteristic zero and also identifies simple module classes with equal characters. In particular, it applies when K is the fraction field of a discrete valuation ring.

3.2 Pullback and quotient descent

For a homomorphism $f : H \rightarrow G$, the pullback $f^* \rho$ acts on the same vector space by $(f^* \rho)(h) = \rho(f(h))$. Its character is $\chi_\rho \circ f$. An intertwiner between two representations of G also intertwines their pullbacks.

Proposition 3.3 (Inflation and descent of representations). *If $f : H \twoheadrightarrow G$ is surjective, then $f^* \rho$ is irreducible if and only if ρ is irreducible. If $\pi : G \twoheadrightarrow H$ and $\ker \pi \leq \ker \rho$, there is a unique representation $\bar{\rho}$ of H on V such that $\pi^* \bar{\rho} = \rho$. It is irreducible whenever ρ is.*

Proof. A subspace invariant under ρ is invariant under its pullback. Conversely, surjectivity lets every $g \in G$ be written as $f(h)$, so an invariant subspace for the pullback is invariant under ρ . The lattices of invariant subspaces are identical.

For descent, define $\bar{\rho}(\pi(g)) = \rho(g)$. If $\pi(g) = \pi(g')$, then $g'^{-1}g \in \ker \pi \leq \ker \rho$, which makes the value independent of the lift. This defines a representation and proves the pullback identity. Surjectivity proves uniqueness, and the first assertion proves irreducibility. $\square$

Implementation.

`ModularRep.Twist` constructs the order isomorphism `Representation.subrepresentationPullbackOrderIso`, from which `Representation.isIrreducible_pullback_iff` follows. These results hold over any field, and the order isomorphism is available more generally over a semiring. In `ModularRep.RepresentationSurjectiveDescent`, descend factors through $G/\ker \pi$ and uses Mathlib's quotient isomorphism $G/\ker \pi \simeq H$. The declarations `descend_apply`, `descend_pullback` and `descend_irreducible` give its defining identities and irreducibility. Uniqueness on the fixed vector space is the direct surjectivity argument in the proof above.

To descend a character, it remains to show that the kernel of the quotient map acts trivially in an affording representation. Over $\mathbb{C}$, the standard identity $\ker \rho = \{g \in G : \chi_\rho(g) = \chi_\rho(1)\}$ gives this directly (see, for example, [5, Lemma 2.19]). The averaging argument below proves the needed implication over any field of characteristic zero, without assuming irreducibility.

Lemma 3.4 (Kernel containment by averaging). *Let S be a finite subgroup of G . If $\chi_\rho(s) = \chi_\rho(1)$ for every $s \in S$, then $S \leq \ker \rho$.*

Proof. The averaging operator

$$P_S = \frac{1}{|S|} \sum_{s \in S} \rho(s)$$

is a projection onto the invariant subspace V^S . Its trace is both $\dim_K V^S$ and, by the hypothesis, $\dim_K V$. These integers are equal because their images in a field of characteristic zero are equal. Hence $V^S = V$, and every element of S acts trivially. $\square$

Implementation.

`Representation.subgroup_le_ker_of_character_eq_one`, in `ModularRep.OrdinaryCharacterLiesOverKernel`, proves this using Mathlib's `Representation.card_inv_mul_sum_char_eq_finrank` for the restricted representation. Only S must be finite. No irreducibility, normality or centrality hypothesis occurs in this lemma.

Theorem 3.5 (Descent of an ordinary irreducible character). *Let $\pi : G \twoheadrightarrow H$ be a surjective homomorphism and let $\chi \in \text{Irr}_K(G)$. Suppose that $\chi(x) = \chi(1)$ for all $x \in \ker \pi$. Then there is a unique $\bar{\chi} \in \text{Irr}_K(H)$ with*

$$\chi(g) = \bar{\chi}(\pi(g)) \quad (g \in G).$$

Proof. Choose an irreducible realisation of χ . Lemma 3.4 places $\ker \pi$ in its kernel. Descend this representation by Proposition 3.3 and take its trace. The resulting representation remains irreducible. Any two functions with the required pullback agree at every element of H , since π is surjective. $\square$

Implementation.

The descended character, its factorisation and its uniqueness are given by `ModularRep.OrdinaryIrreducibleCharacterSurjectiveDescent.descendCharacter`, `descendCharacter_factorisation` and `descendCharacter_unique`. The construction applies averaging to an irreducible realisation and then descends that representation.

For a positive integer a , write a_ℓ for the largest power of the prime ℓ dividing a . An ordinary irreducible character χ has *defect zero at ℓ* if $\chi(1)_\ell = |G|_\ell$, where $\chi(1)$ denotes its positive integral degree. Over a splitting field this is the usual notion of defect zero. Over an arbitrary field of characteristic zero, we use the same numerical condition on the dimension of an irreducible representation in the following statement.

Corollary 3.6 (Preservation of defect zero). *Let ℓ be a prime. Under the hypotheses of Theorem 3.5, assume that $\ell \nmid |\ker \pi|$. If χ has defect zero at ℓ , then $\bar{\chi}$ has defect zero.*

Proof. An affording representation V has dimension $\chi(1)$. Descent retains this vector space and hence the degree. The equality $|G| = |H| |\ker \pi|$ gives $|G|_\ell = |H|_\ell$, so the same dimension satisfies the equality defining defect zero for H . $\square$

Implementation.

`ModularRep.IsDefectZeroOrdinaryCharacter` expresses the dimension equality for an irreducible representation in Mathlib's `FRep` affording the character. `descendCharacter_defectZero` in the descent namespace applies averaging to the affording representation and descends it without changing its dimension.

Example 3.7 (Projection from a direct product). Let $G = H \times C_r$, with $\ell \nmid r$, and let π be the first projection. To apply the descent theorem to an ordinary irreducible character χ , it is enough to verify $\chi(1, c) = \chi(1, 1)$ for every $c \in C_r$. The theorem then constructs $\bar{\chi} \in \text{Irr}_K(H)$ and proves $\chi(h, c) = \bar{\chi}(h)$ for all (h, c) . If the original character has defect zero at ℓ , the corollary preserves that property.

3.3 Linear characters and automorphisms

Write $L_K(G) = \text{Hom}(G, K^\times)$ for the group of linear characters. For $\lambda \in L_K(G)$ define $(\rho \otimes \lambda)(g) = \lambda(g)\rho(g)$ on V . The usual tensor product with a one dimensional representation is realised on V by this formula. For an automorphism α of G , set $\rho^\alpha = \rho \circ \alpha$, $\lambda^\alpha = \lambda \circ \alpha$ and $\chi^\alpha = \chi \circ \alpha$.

Proposition 3.8 (Tensoring with a linear character). *Tensoring with a linear character preserves and reflects irreducibility, and*

$$\chi_{\rho \otimes \lambda}(g) = \lambda(g)\chi_\rho(g).$$

For $\alpha \in \text{Aut}(G)$, one also has

$$(\rho \otimes \lambda)^\alpha = \rho^\alpha \otimes \lambda^\alpha, \quad (\chi \lambda)^\alpha = \chi^\alpha \lambda^\alpha.$$

Proof. Multiplication by the nonzero scalar $\lambda(g)$ does not change which subspaces are invariant under the action of g . Multiplication by $\lambda(g)^{-1}$ gives the converse. The trace formula follows from linearity of trace. The compatibility with α follows by evaluating both sides at g . $\square$

Implementation.

`ModularRep.LinearCharacterTensorAction` defines `Representation.linearCharacterTwist` and proves

`Representation.isIrreducible_linearCharacterTwist_iff`,
`Representation.character_linearCharacterTwist` and
`Representation.linearCharacterTwist_twist`. The induced operation on ordinary irreducible characters is `ModularRep.OrdinaryIrreducibleCharacter.linearTwist`.

The convention for automorphisms is a right action: $(\chi^\alpha)^\beta = \chi^{\alpha\beta}$, with $(\alpha\beta)(g) = \alpha(\beta(g))$. The combined tensor and automorphism action below is instead written as a left action, with the indicated inverses.

More precisely, let A act on G through $a \mapsto \alpha_a$. Its left action on $L_K(G)$ is $a \cdot \lambda = \lambda \circ \alpha_{a^{-1}}$. The resulting semidirect product acts on $\text{Irr}_K(G)$ by

$$(\lambda, a) \cdot \chi = \lambda^{-1} \chi^{\alpha_{a^{-1}}}.$$

The compatibility in Proposition 3.8 proves the action law. The inverses affect formulas for the action, although they do not change the orbits or stabilisers of either factor acting alone.

Implementation.

The declarations

`ModularRep.OrdinaryIrreducibleCharacter.linearCharacterFieldAction`,
`tensorField_semidirectCompatible`, `tensorFieldSemidirectAction` and
`tensorFieldSemidirectAction_apply` specify this convention. They use the semidirect product action in `Formalisation.SemidirectStabilizer`. The right action by automorphisms is represented as a left action of the opposite group. These declarations apply to any group A acting on G .

Example 3.9 (A stabiliser calculation). Suppose A and a subgroup $L \leq L_K(G)$ are given, and A preserves L . Under the combined action, (λ, a) fixes χ if and only if $\chi(\alpha_{a^{-1}}(g)) = \lambda(g)\chi(g)$ for every $g \in G$. Pullback followed by tensoring with a linear character gives an irreducible representation affording the transformed character. To restrict this action to the characters in a block, one must also prove that the block is preserved.

3.4 Extension across a cyclic quotient

Let $N \trianglelefteq H$ with H finite. An extension of a representation ρ of N is a representation $\hat{\rho}$ of H on the same vector space such that $\hat{\rho}|_N \simeq \rho$. An extension makes ρ invariant under conjugation by H : the action of h^{-1} intertwines the conjugation twist of $\hat{\rho}|_N$ with $\hat{\rho}|_N$. Also, an extension of an irreducible representation is irreducible, since every H -invariant subspace is N -invariant.

Theorem 3.10 (Cyclic extension). *Assume that K is algebraically closed. Let H be finite, let $N \trianglelefteq H$, and let ρ be a finite dimensional irreducible K -representation of N . If H/N is cyclic and $\rho \circ c_h \simeq \rho$ for every $h \in H$, where $c_h(n) = hnh^{-1}$, then ρ extends to H .*

Proof. Choose $h \in H$ whose coset generates H/N , and set $r = |H/N|$. Conjugation invariance gives an invertible operator T such that $T\rho(n)T^{-1} = \rho(hnh^{-1})$ for every $n \in N$. Consequently $\rho(h^r)^{-1}T^r$ commutes with $\rho(N)$ and is a nonzero scalar c by Schur's lemma. Choose $u \in K$ with $u^r = c^{-1}$. Replacing T by uT preserves the conjugation identity and gives $T^r = \rho(h^r)$. These identities ensure that $\hat{\rho}(nh^i) = \rho(n)T^i$, for $n \in N$ and $0 \leq i < r$, defines a representation of H extending ρ . $\square$

For the complex character formulation, see [5, Corollary 11.22]. The proof above uses algebraic closure to apply Schur's lemma and take the required root. The modular analogue is discussed in Section 4.7.

Proposition 3.11 (Extension of an invariant ordinary character). *Assume that K is algebraically closed. Let H be finite, let $N \trianglelefteq H$, let H/N be cyclic, and let $\chi \in \text{Irr}_K(N)$. If*

$$\chi(hnh^{-1}) = \chi(n) \quad (h \in H, n \in N),$$

then there exists $\hat{\chi} \in \text{Irr}_K(H)$ such that $\hat{\chi}|_N = \chi$.

Proof. Choose an irreducible realisation ρ of χ . Each conjugation twist of ρ has character χ , so Proposition 3.2 supplies an equivalence with ρ . The cyclic extension theorem gives $\hat{\rho}$ and its restriction equivalence. The extension is irreducible, and taking traces of the intertwining identity proves the required restriction equality. $\square$

Implementation.

`ModularRep.CyclicExtension` defines `Representation.ConjugationInvariant` and `Representation.Extension`. The Lean construction assumes `Representation.CyclicExtensionPrinciple`, which states the representation extension property over the chosen coefficient field. This principle is an explicit hypothesis of the Lean construction, including when K is algebraically closed. Theorem 3.10 proves the mathematical assertion under algebraic closure. Over an arbitrary field of characteristic zero the assertion can fail. In `ModularRep.OrdinaryCharacterCyclicExtensionBridge`, `ModularRep.OrdinaryIrreducibleCharacter.realisation_conjugationInvariant_of_fixed` proves invariance of an irreducible realisation, and `ModularRep.OrdinaryIrreducibleCharacter.exists_extension_of_fixed_cyclic_quotient` constructs the extending character under the stated extension hypothesis. An `ExtensionWitness` consists of an irreducible character of H together with its restriction identity on N .

Example 3.12 (Extension to a semidirect product). Let D and A be finite, with A cyclic. For $H = D \rtimes A$, the normal copy of D has quotient $H/D \simeq A$. A conjugation invariant irreducible representation of D therefore extends to H whenever the cyclic extension property holds over the coefficient field. `Representation.exists_extension_to_semirect_of_cyclic_outer` constructs this quotient isomorphism and applies the principle.

Chapter 4

Brauer characters

Let ℓ be a prime, let G be finite, let k be an algebraically closed field of characteristic ℓ , and let K be a field of characteristic zero. Fix a multiplicative correspondence between the roots of unity in k and K of orders dividing $|G|_{\ell'}$. We use it to lift the eigenvalues of a modular representation and define its Brauer character. For the usual complex valued construction, see, for example, [6, pp. 17–18]. The construction here uses only a finite group of roots of unity in K , which need not be algebraically closed.

We prove additivity on short exact sequences, separation of irreducible representations and linear independence of their Brauer characters. These results will identify composition multiplicities from character values in Chapter 5.

4.1 Regular elements and roots of unity

An element is ℓ -regular if its order is prime to ℓ . Write $G_{\ell'}$ for these elements and $\text{CF}_{\ell'}(G, K)$ for the K -valued functions on $G_{\ell'}$ invariant under conjugation by G . Set

$$m_G = |G|_{\ell'}, \quad \mu_{m_G}(k) = \{a \in k^\times : a^{m_G} = 1\}.$$

Here $|G|_{\ell'}$ is the largest divisor of $|G|$ prime to ℓ . It is a common exponent for the regular elements and need not be their least common exponent.

Definition 4.1. A root correspondence for G is a multiplicative isomorphism $\iota_G : \mu_{m_G}(k) \rightarrow \mu_{m_G}(K)$.

Since k is algebraically closed and $\ell \nmid m_G$, its group of m_G th roots is cyclic of order m_G . The existence of ι_G therefore requires K to contain all m_G th roots of unity. The correspondence ι_G is specified separately from the modular system in Chapter 6. When restricting characters or comparing character tables, the chosen correspondences must agree on the roots involved.

Implementation.

`ModularRep.BrauerRootConvention` defines `PrimeRegularRootEmbedding`. `AgreesOnRoots`: two correspondences agree on every root whose order divides the exponent for the group being restricted to. Agreement is reflexive and is transitive when the required exponents divide one another. The theorem `agreesOnRoots_of_commonRoot` proves agreement for correspondences induced by one common root identification. Separate choices of roots do not establish this agreement.

Lemma 4.2 (Orders of eigenvalues at regular elements). *Let ρ be a finite dimensional k -representation of G . If $g \in G_{\ell'}$ and a is a root of the characteristic polynomial of $\rho(g)$, then $a^{m_G} = 1$.*

Proof. The order of g divides $|G|$ and is prime to ℓ , so it divides m_G . Hence $g^{m_G} = 1$. A root of the characteristic polynomial is an eigenvalue, so there is $v \neq 0$ with $\rho(g)v = av$. Applying the m_G th power gives $v = a^{m_G}v$, and cancellation proves the assertion. $\square$

Implementation.

`ModularRep.primeRegularExponent` is m_G . `ModularRep.PrimeRegularRootEmbedding` specifies a prime p and an equivalence between the groups defined by Mathlib's `rootsOfUnity`. The exponent identity and the eigenvalue lemma are `ModularRep.PrimeRegularElement.pow_primeRegularExponent_eq_one` and `Representation.charpolyRoot_pow_primeRegularExponent_eq_one` in `ModularRep.PrimeRegularEigenvalues`. These two elementary results only need a field and a finite dimensional representation. The field characteristic and algebraic closure enter the subsequent construction of Brauer characters.

4.2 The character of a modular representation

For a finite dimensional k -representation ρ , let $\mathcal{R}_\rho(g)$ be the multiset of roots of the characteristic polynomial of $\rho(g)$, with algebraic multiplicities.

Definition 4.3. The Brauer character afforded by ρ , relative to ι_G , is

$$\varphi_\rho(g) = \sum_{a \in \mathcal{R}_\rho(g)} \iota_G(a) \quad (g \in G_{\ell'}).$$

An irreducible Brauer character is one afforded by an irreducible finite dimensional k -representation. Their set is denoted $\text{IBr}_{\iota_G}(G)$.

The preceding lemma places every root in the domain of ι_G . Algebraic closure ensures that the characteristic polynomial splits over k . The eigenvalues are lifted to K before they are summed. Their sum in k is the trace $\text{tr}_k(\rho(g))$.

Proposition 4.4 (Invariance of Brauer characters). *The function φ_ρ is constant on conjugacy classes in $G_{\ell'}$. Equivalent representations have equal Brauer characters. For $\alpha \in \text{Aut}(G)$,*

$$\varphi_{\rho \circ \alpha}(g) = \varphi_\rho(\alpha(g)).$$

Proof. Conjugating a group element conjugates its representing operator. An equivalence of representations also conjugates the corresponding operators. Conjugate operators have the same characteristic polynomial, hence the same root multiset. The definition as a sum of lifted roots gives the first two assertions. For the last assertion the operator on the left at g is exactly the operator on the right at $\alpha(g)$. $\square$

Implementation.

`ModularRep.BrauerCharacter` defines `Representation.brauerCharacterOfRootEmbedding` using Mathlib's characteristic polynomials and their root multisets. It has values in `ModularRep.PrimeRegularClassFunction`. It uses a total function `PrimeRegularRootEmbedding.lift`, extended by zero outside the root group, to map the root multiset. The theorem `Representation.lift_charpolyRootAsRootOfUnity` proves agreement with the specified root equivalence on every root that occurs. The invariance declarations are `Representation.charpoly_conj`, `Representation.brauerCharacterOfRootEmbedding_iso` and `Representation.brauerCharacterOfRootEmbedding_twist`. The conjugacy lemma specialises Mathlib's `LinearEquiv.charpoly_conj` to representation operators. `ModularRep.IBr` consists of class functions admitting an irreducible FDRP realisation. Equality of its elements is pointwise equality of functions.

Example 4.5 (Diagonal representations). Suppose $g \in G_{\ell'}$ acts diagonally with entries $a_1, \dots, a_d \in \mu_{m_G}(k)$. Its Brauer value is $\iota_G(a_1) + \dots + \iota_G(a_d)$. If several entries coincide, every occurrence contributes to the sum. At the identity all entries are one, so $\varphi_\rho(1) = d$ in K . This remains the integer d even when the trace of the identity in k vanishes because $\ell \mid d$.

4.3 Additivity on exact sequences

Lemma 4.6 (Characteristic polynomial on a subspace and quotient). *Let T be an endomorphism of a finite dimensional vector space V over a field, and let $U \leq V$ be T -invariant. If $\bar{T}$ is the induced endomorphism of V/U , then*

$$\det(XI - T) = \det(XI - T|_U) \det(XI - \bar{T}).$$

Proof. Choose a basis of U , a basis of V/U and lifts of the quotient basis. Together they give a basis of V in which T has the form

$$\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}.$$

The diagonal blocks represent $T|_U$ and $\bar{T}$. The determinant of the resulting block upper triangular polynomial matrix is the product of the two diagonal determinants. $\square$

Implementation.

`LinearMap.charpoly_eq_restrict_mul_mapQ` in
`ModularRep.CharpolyInvariantSubmodule` constructs this basis and identifies all four blocks. It uses Mathlib's basis construction `Module.Basis.sumQuot` and block matrix identity `Matrix.charpoly_fromBlocks_zero21`.

Theorem 4.7 (Additivity on short exact sequences). *For a short exact sequence of finite dimensional kG -representations*

$$0 \longrightarrow U \xrightarrow{f} V \xrightarrow{\pi} W \longrightarrow 0,$$

the Brauer characters formed using ι_G satisfy $\varphi_V = \varphi_U + \varphi_W$. In particular, $\varphi_{U \oplus W} = \varphi_U + \varphi_W$.

Proof. First take an invariant subspace of V . By Lemma 4.6, the root multiset of the operator on V is the sum of the root multisets on the subspace and its quotient. Applying the same lift to each occurrence and summing proves additivity. This step uses no additive property of ι_G . For a short exact sequence, $\text{im } f = \ker \pi$ is invariant. Injectivity of f identifies U with $\text{im } f$, and surjectivity of π identifies $V/\text{im } f$ with W . Both identifications intertwine the actions. The first part and Proposition 4.4 give the assertion. The case of a direct sum follows in the same way, or from the characteristic polynomial of a block diagonal operator. $\square$

Implementation.

`ModularRep.BrauerCharacterExact` proves
`Representation.brauerCharacterOfRootEmbedding_subrepresentation_quotient` and
`Representation.brauerCharacterOfRootEmbedding_prod`. The statement for the category of finite dimensional representations is `FDRep.brauerCharacterOfRootEmbedding_shortExact`. `FDRep.shortExact_underlying_functions` gives injectivity, surjectivity and exactness of the underlying linear maps. Intertwining identities identify the subrepresentation and quotient with the first and last terms of the sequence.

Example 4.8 (An invariant filtration). If $0 = V_0 \leq V_1 \leq \dots \leq V_r = V$ is a finite invariant filtration, repeated exact additivity gives $\varphi_V = \sum_{i=1}^r \varphi_{V_i/V_{i-1}}$. For a representation given by

block upper triangular matrices, the diagonal blocks therefore suffice to calculate its Brauer character. The blocks above the diagonal can distinguish a nonsplit extension from a direct sum, even though their Brauer characters agree.

4.4 Separation and linear independence

The modular traces of nonisomorphic simple representations are linearly independent. To pass from this fact to Brauer characters, we compare the same sums of roots of unity in the two fields. Cyclotomic polynomials provide the comparison. For linear independence of complex valued Brauer characters, see [6, Theorem 2.6].

Lemma 4.9 (Traces on regular parts). *Let ρ be a finite dimensional k -representation of G . For every $g \in G$, write $g = g_\ell g_{\ell'}$ as commuting factors of ℓ -power order and order prime to ℓ . Then*

$$\mathrm{tr}_k(\rho(g)) = \mathrm{tr}_k(\rho(g_{\ell'})).$$

This trace identity also appears in [6, Lemma 2.4].

Proof. If g_ℓ has order dividing ℓ^a , then in characteristic ℓ the operator $P = \rho(g_\ell)$ satisfies $(P - I)^{\ell^a} = P^{\ell^a} - I = 0$. Let $R = \rho(g_{\ell'})$. The operators P and R commute. Hence $R(P - I)$ is nilpotent and has trace zero. Since $RP = R + R(P - I)$, taking traces proves the claim. $\square$

Implementation.

`Representation.character_eq_character_primeRegularPart` in `ModularRep.ModularTraceRegularPart` uses the commuting primary decomposition from `ModularRep.PrimeRegularPart` and Mathlib's trace identity for a commuting nilpotent term.

Proposition 4.10 (Brauer characters determine irreducible representations). *If two finite dimensional modular representations have equal Brauer characters relative to ι_G , then their k -valued trace functions on G are equal. If both representations are irreducible, they are equivalent.*

Proof. Choose a primitive m_G th root $\zeta \in k$. Its image $\hat{\zeta} = \iota_G(\zeta)$ is primitive in K . At a regular element, write the two eigenvalue multisets as powers of ζ . The difference of the sums of these powers is the value of some polynomial $F \in \mathbb{Z}[X]$ at ζ . Equality of the Brauer values says $F(\hat{\zeta}) = 0$. The minimal polynomial of $\hat{\zeta}$ over $\mathbb{Q}$ is the cyclotomic polynomial Φ_{m_G} , so Φ_{m_G} divides F in $\mathbb{Z}[X]$. Since m_G is prime to ℓ , the primitive root ζ is also a root of the corresponding cyclotomic polynomial in k . Thus $F(\zeta) = 0$.

The sum of the roots of the characteristic polynomial is the trace, so the traces agree at regular elements. Lemma 4.9 gives agreement on all of G . The argument separating simple modules by their traces in Proposition 3.2, applied over the algebraically closed field k , proves the last assertion. $\square$

Implementation.

`ModularRep.CyclotomicRootSumReflection` proves that equality of the lifted sums in K implies equality of the sums in k . `ModularRep.BrauerTraceRecovery` identifies these sums with the root multisets used by the character definition. `ModularRep.FDRepSimpleClassKZero.brauerCharacterDeterminesModularTrace_of_rootEmbedding` and `ModularRep.FDRepSimpleClassKZero.irreducibleBrauerCharacterInjectivity_of_rootEmbedding` are the resulting theorems in `ModularRep.BrauerCharacterSeparation`. The latter proves that simple modules with equal Brauer characters are isomorphic. Trace recovery is proved from the root correspondence.

Theorem 4.11 (Linear independence of irreducible Brauer characters). *The functions in $\text{IBr}_{\iota_G}(G)$ are linearly independent over K on G_ℓ .*

Proof. Choose one representative of every simple kG -module class. There are finitely many such classes: each is a composition factor of the finite dimensional regular module. Denote the representatives by $S_1, \dots, S_s$ and their trace functions by $t_1, \dots, t_s$.

First, the t_i are linearly independent over k . For each i , Schur's lemma and density on $\bigoplus_j S_j$ give an element of kG acting as an operator of trace one on S_i and as zero on every other summand. Evaluating any linear relation of trace functionals at this element extracts its i th coefficient. Restriction of a linear functional on kG to the basis G is injective, so the trace functions on G are independent too. Lemma 4.9 then gives independence of their restrictions to G_ℓ .

Independent functions admit a nonsingular square evaluation matrix. Indeed, their evaluation vectors span k^s , since a linear functional annihilating that span would give a relation between the functions. Choose a basis of evaluation vectors, at regular elements $x_1, \dots, x_s$, and write

$$M_k = (t_i(x_j))_{i,j}, \quad \det M_k \neq 0.$$

Express each root multiset occurring in this matrix as powers of the primitive root ζ . Replacing each ζ by the class of X defines a matrix M over the common ring $R = \mathbb{Z}[X]/(\Phi_{m_G})$. Evaluation at ζ gives M_k . Evaluation at $\hat{\zeta}$ gives the Brauer evaluation matrix $M_K = (\varphi_{S_i}(x_j))_{i,j}$.

The map $R \rightarrow K$ is injective by the theorem on the minimal polynomial of a primitive root of unity. If $\det M_K = 0$, then $\det M = 0$ in R , and evaluating in k would give $\det M_k = 0$. This contradiction shows $\det M_K \neq 0$. Evaluating a K -linear relation among the Brauer characters at $x_1, \dots, x_s$ now forces all coefficients to vanish. Finally, separation and realisation identify this family indexed by simple classes with the set of functions $\text{IBr}_{\iota_G}(G)$. $\square$

Implementation.

`ModularRep.SimpleTraceLinearIndependence` proves independence of the modular simple trace functions, and `ModularRep.PrimeRegularEvaluationMatrix` obtains the regular evaluation points. The coefficient argument is `ModularRep.CyclotomicDeterminantSpecialization.rootMultisetMatrix_det_ne_zero_transfer`, using `CyclotomicOrder` and `cyclotomicSpecialization_injective` in the same namespace. `ModularRep.FDRepSimpleClassKZero.exists_simpleClass_primeRegular_brauer_det_ne_zero` constructs the nonsingular Brauer evaluation matrix. The final theorem is `ModularRep.FDRepSimpleClassKZero.irreducibleBrauerCharacters_linearIndependent`. The determinant is first computed over the cyclotomic integer ring. Its nonvanishing after evaluation in K permits the final linear algebra argument over K .

Example 4.12 (Equality of composition multiplicities). Suppose exact additivity gives $\varphi_V = \sum_i a_i \varphi_i$ and $\varphi_W = \sum_i b_i \varphi_i$, where a_i, b_i are nonnegative integers and the φ_i are distinct irreducible Brauer characters. If $\varphi_V = \varphi_W$, linear independence gives $a_i = b_i$ for every i , using characteristic zero to recover equality of integers from equality in K . This is the step used to prove injectivity of the Brauer character map on the exact Grothendieck group. It determines composition multiplicities without asserting that $V \simeq W$.

4.5 Compatible choices under pullback and restriction

Let $f : H \rightarrow G$ be a homomorphism of finite groups with root correspondences ι_H and ι_G . A regular element of H maps to a regular element of G , so class functions can always be pulled

back by f . To compare this pullback with the character of $f^*\rho$, the two lifts must agree on the roots of $\rho(f(h))$ for every $h \in H_{\ell'}$.

Proposition 4.13 (Pullback with compatible root correspondences). *Let $f : H \rightarrow G$ be a homomorphism and let ρ be a finite dimensional k -representation of G . Suppose that $\iota_H(a) = \iota_G(a)$ for every eigenvalue a of $\rho(f(h))$, for every $h \in H_{\ell'}$. Then*

$$\varphi_{f^*\rho}^{\iota_H}(h) = \varphi_{\rho}^{\iota_G}(f(h)) \quad (h \in H_{\ell'}).$$

Such agreement holds if the two correspondences are restrictions of one multiplicative equivalence $\mu_M(k) \simeq \mu_M(K)$, where both m_H and m_G divide M .

Proof. The representing operator on both sides is $\rho(f(h))$. Its root multiset is therefore the same on both sides, and the agreement of the lifts gives equality term by term. A common root correspondence gives that agreement on every root in the intersection of the two root groups. $\square$

Implementation.

Compatibility on these eigenvalues is expressed by

`Representation.BrauerRootLiftCompatibleAlong`.

`Representation.brauerCharacterOfRootEmbedding_pullback_of_compatible` proves the formula. `ModularRep.BrauerCharacterCommonRootCompatibility` restricts one common equivalence and proves the condition for any homomorphism. Equality of the total functions $k \rightarrow K$ is a sufficient stronger condition, but is unnecessary away from the occurring roots.

A group isomorphism $\phi : G \rightarrow H$ preserves $|G|_{\ell'}$ and therefore transports a chosen root correspondence with the same function lifting roots. Pullback along ϕ^{-1} then transports an irreducible Brauer character together with an affording representation. The constructions `PrimeRegularRootEmbedding.alongMulEquiv` and `IrreducibleBrauerCharacter.alongMulEquiv` in `ModularRep.BrauerCharacterEquivTransport` give this transport. If an extension is also transported, the ambient and base group isomorphisms must commute with the inclusion maps. A separately specified target character must be identified with the transported base character. These are the hypotheses of `Representation.Extension.BrauerCharacterExtensionWitness.alongMulEquivOfBase` in `ModularRep.BrauerCharacterExtensionWitnessTransport`.

Proposition 4.14 (Extending a root correspondence from a subgroup). *Let $H \leq G$, and prescribe a root correspondence for H . If at least one root correspondence for G exists, one can choose a correspondence for G whose restriction agrees with the prescribed correspondence on $\mu_{m_H}(k)$.*

Proof. The divisibility $m_H \mid m_G$ permits restriction of any chosen ambient correspondence to the smaller root group. If j is this restriction, then $j^{-1}\iota_H$ is an automorphism of the finite cyclic group $\mu_{m_H}(k)$. An automorphism of a finite cyclic subgroup extends to the ambient finite cyclic group: in power coordinates this is surjectivity of reduction on the corresponding groups of units. Compose the ambient correspondence with such an extension. Its restriction is now the prescribed one. $\square$

Implementation.

`ModularRep.PrimeRegularRootEmbedding.exists_ambient_agreeing_on_subgroup`

assumes existence of an ambient root correspondence, expressed by `Nonempty`, and uses `exists_mulAut_extension` in the same namespace. It adjusts an ambient correspondence while retaining the prescribed subgroup correspondence.

There is a particularly direct choice for a normal ℓ -subgroup $N \trianglelefteq G$. Since $|G| = |G/N| |N|$, one has $m_G = m_{G/N}$. A prescribed quotient correspondence can therefore be transported to

G with the same root lifting map. `ModularRep.PrimeRegularRootEmbeddingPQuotient.exponent_quotient_eq`, `ofPQuotient` and `ofPQuotient_lift` implement this observation.

4.6 Inflation and descent of irreducible Brauer characters

For $\varphi \in \text{IBr}_{\iota_G}(G)$, define $\ker \varphi = \ker \rho$, where ρ is any irreducible representation affording φ . Proposition 4.10 makes this independent of the choice of ρ , since equivalent representations have the same kernel. Let $\pi : G \rightarrow H$ be a surjection.

Theorem 4.15 (Inflation of irreducible Brauer characters). *Suppose that $\ell \nmid |\ker \pi|$, and that the root correspondences are compatible along π for every finite dimensional representation of H . Inflation gives a bijection between $\text{IBr}_{\iota_H}(H)$ and $\{\varphi \in \text{IBr}_{\iota_G}(G) : \ker \pi \leq \ker \varphi\}$. If $\bar{\varphi}$ corresponds to φ , then*

$$\varphi(g) = \bar{\varphi}(\pi(g)) \quad (g \in G_{\ell'}).$$

Proof. Pullback along π preserves irreducibility, and the pulled back representation has $\ker \pi$ in its kernel. Root compatibility identifies its Brauer character with the inflated function. Conversely, descend a representation affording φ on the same vector space, using $\ker \pi \leq \ker \varphi$. Representation descent preserves irreducibility, and compatible pullback gives the asserted identity.

It remains to prove that pullback is injective on functions on $H_{\ell'}$. Every lift g of an ℓ -regular h is itself ℓ -regular in this case. Indeed, if h has order d , then $g^d \in \ker \pi$, so the order of g divides $d |\ker \pi|$, which is prime to ℓ . Hence π is surjective on regular elements and pullback of their class functions is injective. The two character identities now show that inflation and descent are inverse. $\square$

Implementation.

`ModularRep.IrreducibleBrauerCharacterSurjectiveDescent.KernelTrivialIBrAlong` consists of irreducible Brauer characters admitting a realisation on which $\ker \pi$ acts trivially. By character separation, this is the condition $\ker \pi \leq \ker \varphi$. In the same namespace, `inflateToKernelTrivialIBrAlong`, `descendIBrAlong` and `quotientIBrEquivKernelTrivialIBrAlong` construct the maps and the bijection. The representation descent itself only requires surjectivity and an irreducible realisation on which $\ker \pi$ acts trivially. Root compatibility proves its character identity. The hypothesis that $\ker \pi$ has order prime to ℓ proves injectivity of pullback on class functions. This hypothesis differs from the normal ℓ -subgroup hypothesis in the preceding construction of a root correspondence. The classical inflation correspondence allows an arbitrary normal kernel [6, p. 39]. The theorem stated here uses the stronger hypothesis that the kernel has order prime to ℓ , as does the cited Lean equivalence.

4.7 Tensor actions and cyclic extensions

For a modular linear character $\lambda : G \rightarrow k^\times$, define $\widehat{\lambda}(g) = \iota_G(\lambda(g))$ on $G_{\ell'}$. The value $\lambda(g)$ belongs to the required root group because $g^{m_G} = 1$. Multiplicativity of the root correspondence proves $\widehat{\lambda\mu} = \widehat{\lambda}\widehat{\mu}$ and $(\widehat{\lambda})^\alpha = \widehat{\lambda \circ \alpha}$.

Theorem 4.16 (Tensoring with a linear character). *For every finite dimensional modular representation V and every $\lambda : G \rightarrow k^\times$, one has*

$$\varphi_{V \otimes \lambda} = \widehat{\lambda} \varphi_V.$$

Proof. At $g \in G_{\ell'}$, the representing operator on $V \otimes \lambda$ is $\lambda(g)\rho_V(g)$. Its eigenvalues, counted with multiplicity, are $\lambda(g)a$ as a runs through the eigenvalues of $\rho_V(g)$. Multiplicativity of ι_G gives $\iota_G(\lambda(g)a) = \widehat{\lambda}(g)\iota_G(a)$. Summing proves the formula. $\square$

This is the case of a one dimensional factor in the tensor product formula for Brauer characters. See, for example, [6, proof of Theorem 2.23]. The argument with invariant subspaces in Proposition 3.8 also applies over k . Thus $V \otimes \lambda$ is irreducible when V is irreducible, and $\widehat{\lambda}\varphi_V$ is then an irreducible Brauer character.

Implementation.

The Lean construction uses `ModularRep.BrauerLinearTensorProductFormula` as a hypothesis. `ModularRep.BrauerLinearCharacterTensorAction` defines `ModularRep.PrimeRegularRootEmbedding.liftedLinearCharacter` and proves its multiplicative and automorphism identities. Given the tensor formula, `ModularRep.IrreducibleBrauerCharacter.linearTwist` constructs the irreducible Brauer character. Its `linearTwist_mul`, `twist_linearTwist` and `tensorFieldSemidirectAction` give the action laws, with the inverse convention explained in Section 3.3.

Theorem 4.17 (Cyclic extension of a Brauer character). *Let H be finite, let $N \trianglelefteq H$, and suppose that H/N is cyclic. Choose root correspondences ι_N and ι_H compatible with restriction to N for every finite dimensional k -representation of H . If $\varphi \in \text{IBr}_{\iota_N}(N)$ is invariant under conjugation by H , then there is $\widehat{\varphi} \in \text{IBr}_{\iota_H}(H)$ with $\widehat{\varphi}|_{N_{\ell'}} = \varphi$.*

Proof. Choose an irreducible realisation ρ of φ . Its conjugation twists have the same Brauer character, so Proposition 4.10 makes them equivalent to ρ . The proof of Theorem 3.10 applies over the algebraically closed field k in any characteristic. It therefore gives an extension $\widehat{\rho}$, which is irreducible because ρ is irreducible. Compatible roots identify the restriction of its Brauer character with φ by Proposition 4.13. $\square$

For the usual complex valued formulation and the representation extension argument, see [6, Theorem 8.12]. There is no restriction on the order of the cyclic quotient relative to ℓ . The value field K need not be algebraically closed, but the root correspondences must satisfy the stated compatibility.

Implementation.

The Lean construction assumes `Representation.BrauerCyclicExtensionPrinciple` over the algebraically closed modular field k . It does not prove this extension property from algebraic closure. `ModularRep.BrauerCharacterExtensionBridge` proves `Representation.conjugationInvariant_of_brauerCharacter_fixed` and `Representation.exists_extension_realisation_of_ibr_fixed_cyclic_quotient`. The latter gives an affording representation and a representation extension. In `ModularRep.BrauerCharacterHomPullback`, `Representation.Extension.brauerCharacterExtensionWitnessOfCompatible` uses compatible roots to construct the extending irreducible Brauer character and its restriction equality.

Chapter 5

Grothendieck groups and composition factors

The Grothendieck group assigns a class $[V]$ to each representation, with $[V] = [U] + [W]$ for every short exact sequence $0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$. For finite dimensional representations, the Jordan–Hölder theorem shows that the classes of simple modules form a basis. Thus composition multiplicities determine the class of a representation.

In the parts of this chapter concerning representations, G is a finite group and k is a field. No splitting or characteristic assumption is needed until the final section on Brauer characters. Write kG for the group algebra, $\text{FDRep}_k(G)$ for the category of finite dimensional k -representations, and $K_0(kG)$ for its Grothendieck group defined by short exact sequences.

5.1 The Grothendieck group and its universal property

Definition 5.1. Let $\mathcal{C}$ be an abelian category whose isomorphism classes of objects form a set. Let $F(\mathcal{C})$ be the free abelian group on the isomorphism classes of its objects. The Grothendieck group defined by short exact sequences is

$$K_0(\mathcal{C}) = F(\mathcal{C})/R(\mathcal{C}),$$

where

$$R(\mathcal{C}) = \langle e_V - e_U - e_W : 0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0 \text{ is exact} \rangle.$$

The symbol e_X denotes the generator corresponding to the isomorphism class of X . Its image in the quotient is denoted by $[X]$.

For categories of modules, this definition is recalled in [8, Appendix, p. 163]. Isomorphic objects have the same class. Each defining relation gives $[V] = [U] + [W]$. The short exact sequence with all three terms zero implies $[0] = [0] + [0]$, hence $[0] = 0$. The split sequence for $U \oplus W$ gives $[U \oplus W] = [U] + [W]$. Exact relations also apply to extensions that do not split, which is the feature needed for modular representations.

Proposition 5.2 (Universal property). *Let A be an abelian group. An isomorphism invariant assignment $X \mapsto a_X$ defines a homomorphism $K_0(\mathcal{C}) \rightarrow A$ with $[X] \mapsto a_X$ if and only if*

$$a_V = a_U + a_W$$

for every short exact sequence $0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$. Such a homomorphism is unique. Consequently two homomorphisms from $K_0(\mathcal{C})$ agree if they agree on the classes $[X]$.

Proof. The assignment defines a unique homomorphism $F(\mathcal{C}) \rightarrow A$ by the universal property of the free abelian group. It vanishes on $e_V - e_U - e_W$ precisely when the stated identity holds. Vanishing on the generators of $R(\mathcal{C})$ is equivalent to vanishing on that subgroup, so the homomorphism descends to the quotient. Conversely every homomorphism on the quotient respects its defining relations. Object classes generate the quotient, which proves uniqueness. $\square$

Implementation.

`ModularRep.ExactGrothendieckGroup` constructs this presentation from Mathlib's skeleton, free abelian group and short exact complexes. Here `FreeClassGroup` denotes the free abelian group, `shortExactRelationSubgroup` its subgroup of relations, and `KZero` the quotient. The identities for isomorphic objects and short exact sequences are `classOf_iso` and `classOf_middle_eq`. The universal property is `lift`, with `lift_classOf` giving its value on an object class and `hom_ext` expressing uniqueness. The elements of `Skeleton C` are isomorphism classes of objects. These definitions also apply to categories with zero morphisms, using short exact complexes $U \rightarrow V \rightarrow W$. The statements above concern abelian categories.

Proposition 5.3 (Maps induced by exact functors). *Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an exact functor between abelian categories, that is, an additive functor that preserves short exact sequences. Then F induces a homomorphism*

$$K_0(F) : K_0(\mathcal{C}) \longrightarrow K_0(\mathcal{D}), \quad [X] \longmapsto [F(X)].$$

Proof. The hypotheses ensure that the image of a short exact sequence is short exact. Thus the free homomorphism induced by F sends every defining relation to a defining relation for $K_0(\mathcal{D})$. It therefore induces a homomorphism on the quotients. $\square$

Implementation.

The induced homomorphism is `ExactGrothendieckGroup.map`. Its hypotheses require preservation of zero morphisms, finite limits and finite colimits. For abelian categories, these are equivalent to exactness. The construction uses `freeMap_shortExactRelation` and `map_shortExactRelationSubgroup`. The formula on object classes is `map_classOf`. No choice of bases is involved.

5.2 Composition factors and the Jordan–Hölder theorem

Let R be a ring and M a left R -module of finite length. A composition series of M is a chain

$$0 = M_0 < M_1 < \cdots < M_r = M$$

whose successive quotients M_{i+1}/M_i are simple. We also consider composition series between two fixed submodules $A \leq B$ of M , with first term A and last term B . For an account of composition series and the Jordan–Hölder theorem, see, for example, [2, §1.1, especially Theorem 1.1.4].

Definition 5.4. Write $\text{Mod } R$ for the category of all left R -modules in the chosen universe. Recall that $F(\text{Mod } R)$ is the free abelian group on their isomorphism classes. For a composition series s of M , its formal factor sum is

$$J(s) = \sum_{i=0}^{r-1} e_{M_{i+1}/M_i} \in F(\text{Mod } R),$$

and the coefficient $m_X(s)$ at an isomorphism class X counts the factors isomorphic to X .

Proposition 5.5 (Independence of composition multiplicities). *Any two composition series s and t between the same two submodules of M have the same simple factors up to isomorphism, counted with multiplicity. Thus $J(s) = J(t)$ and $m_X(s) = m_X(t)$ for every module isomorphism class X . In particular, a composition series from 0 to a module of finite length defines $J(M)$ and $m_X(M)$ independently of the choice of series.*

Proof. The Jordan–Hölder theorem gives a bijection between the factor indices of the two series and an isomorphism between corresponding factors. After passing to isomorphism classes, reindexing the finite sum gives the result. This proves equality in the free abelian group, before imposing exact relations. $\square$

Implementation.

`ModularRep.JordanHolderCoordinates` defines `compositionFactor`, `compositionFactorIsoSum`, and `compositionCoefficient`. Each quotient is formed inside its numerator: the denominator in `compositionFactor` is the inverse image of the preceding submodule under the numerator’s inclusion into M . Simplicity follows from the covering relation between adjacent submodules in `compositionFactor_isSimple`. The proof of `compositionFactorIsoSum_eq` applies Mathlib’s `CompositionSeries.jordan_holder` and the corresponding module isomorphisms. The formula `compositionCoefficient_eq_sum_ite` counts the factors in each isomorphism class.

Every finite dimensional representation has finite length as a module over its monoid algebra, even when the monoid is infinite. Indeed it is both Noetherian and Artinian as a vector space. A chain of submodules over the monoid algebra is also a chain of vector subspaces, so both chain conditions pass to that module structure. In `ModularRep.FDRepFiniteLength`, `asModule_isFiniteLength` applies Mathlib’s criteria for finite length and its results on restriction of scalars. The theorem `exists_compositionSeries` then applies Mathlib’s equivalence between finite length and the existence of a composition series from 0 to the whole module. For finite G , all simple kG -modules are finite dimensional, as shown in Section 5.4.

5.3 Splicing series in a short exact sequence

Proposition 5.6 (Additivity of composition factors). *For a short exact sequence of finite length left R -modules*

$$0 \longrightarrow U \xrightarrow{f} V \xrightarrow{\pi} W \longrightarrow 0,$$

the formal factor sums satisfy $J(V) = J(U) + J(W)$. Consequently $m_X(V) = m_X(U) + m_X(W)$ for every module isomorphism class X .

Proof. Choose composition series $0 = U_0 < \cdots < U_a = U$ and $0 = W_0 < \cdots < W_b = W$. The images $f(U_i)$ form a composition series from 0 to $\text{im } f$, since f is injective. The inverse images $\pi^{-1}(W_j)$ form one from $\ker \pi$ to V , since π is surjective. Exactness identifies the meeting terms $\text{im } f = \ker \pi$, so the two chains concatenate to

$$0 = f(U_0) < \cdots < f(U_a) = \pi^{-1}(W_0) < \cdots < \pi^{-1}(W_b) = V.$$

The factor isomorphisms are

$$U_{i+1}/U_i \simeq f(U_{i+1})/f(U_i), \quad \pi^{-1}(W_{j+1})/\pi^{-1}(W_j) \simeq W_{j+1}/W_j.$$

The first is induced by the injective map f . For the second, restrict π to $\pi^{-1}(W_{j+1})$. The induced map on quotients is surjective and has zero kernel, since the inverse image of W_j is $\pi^{-1}(W_j)$. Thus the concatenated series has factor sum $J(U) + J(W)$. Jordan–Hölder compares it with any chosen full series of V . $\square$

Implementation.

The proof in `ModularRep.FDRepJordanHolderKZero` constructs these chains as `mapCompositionSeries` and `comapCompositionSeries`. The factor isomorphisms are `quotientMapEquivOfInjective` and `quotientComapEquivOfSurjective`. Concatenation uses Mathlib's composition series operation `smash`, and the local theorem `compositionFactorIsoSum_smash` expresses its factor sum as the sum of the two factor sums. This gives `compositionFactorIsoSum_eq_add_of_exact`.

For representations, `fdRepJordanHolderSum_shortExact` applies the exact underlying kG -module functor and uses the injection, surjection, and exactness of the resulting linear maps. The universal property gives `fdRepJordanHolderKZeroHom`, with values in the free abelian group on module isomorphism classes.

5.4 The basis of simple modules

Let $\mathcal{S}_k(G)$ denote the set of isomorphism classes of simple left kG -modules. Every such module is cyclic: any nonzero vector generates a nonzero submodule and hence the whole module. Since G is finite, kG is finite dimensional over k , so every simple module belongs to $\text{FDRep}_k(G)$.

For an account of the following standard description of the Grothendieck group, see [8, §14.1, Proposition 40].

Theorem 5.7 (Basis of simple module classes). *The map*

$$K_0(kG) \longrightarrow \bigoplus_{S \in \mathcal{S}_k(G)} \mathbb{Z}e_S, \quad [V] \longmapsto \sum_S m_S(V)e_S$$

is an isomorphism of abelian groups. Its inverse sends e_S to $[S]$.

Proof. All factors in $J(V)$ are simple, so its sum lies in the subgroup freely generated by $\mathcal{S}_k(G)$. The additivity just proved gives the homomorphism in the statement. In the other direction, the free abelian group admits a homomorphism $e_S \mapsto [S]$.

For a composition series from 0 to V , each step has a short exact sequence

$$0 \longrightarrow V_i \longrightarrow V_{i+1} \longrightarrow V_{i+1}/V_i \longrightarrow 0.$$

Starting from $[0] = 0$ and successively applying the defining relations gives

$$[V] = \sum_{i=0}^{r-1} [V_{i+1}/V_i].$$

Thus the composite back to $K_0(kG)$ fixes every object class and hence the whole group. For a simple module S , the series $0 < S$ has the single factor S . The other composite therefore fixes every generator e_S . The maps are inverse. $\square$

Implementation.

In `ModularRep.FDRepSimpleClassKZero`, `SimpleModuleClass` is the subtype of the module skeleton represented by simple modules. The homomorphism `simpleClassProjection` fixes the generators represented by simple modules and sends the others to zero. Composing it with the homomorphism defined by composition multiplicities gives `fdRepSimpleJordanHolderHom`.

The reverse map is `simpleClassToFDRepKZero`. The proof first sums the step relations in the category `FGModuleCat k[G]` of finitely generated kG -modules using

`compositionSeriesFGKZero_decomposition`, and then applies `FDRepGroupAlgebraEquivalence.equivalenceFGModule`. This is a restriction of Mathlib's equivalence between representations and modules over the monoid algebra. Finiteness of G ensures that finite generation over kG implies finite dimension over k . The resulting isomorphism is `fdRepKZeroEquivSimpleModuleClassGroup`.

Example 5.8 (Classes of extensions of simple modules). Suppose $0 \rightarrow S \rightarrow V \rightarrow T \rightarrow 0$ is short exact, where S and T are simple. Then $[V] = [S] + [T]$. If $S \not\simeq T$, the coefficients of $[S]$ and $[T]$ are one and all other coefficients are zero. If $S \simeq T$, the coefficient of $[S]$ is two and all other coefficients are zero. Thus any two such extensions have the same class in $K_0(kG)$, even when their middle terms are not isomorphic.

For virtual classes, coefficients may be negative. Equality in $K_0(kG)$ means equality of all these integer coefficients. For a representation V , the coefficient of $[S]$ is $m_S(V)$, its composition multiplicity. This is the identity `FDRepJordanHolderKZero.fDRepJordanHolderCoordinate_classOf`.

5.5 Finiteness of the set of simple classes

Proposition 5.9 (Finiteness of the simple module classes). *If the left regular module of a ring R has a full finite composition series, every simple left R -module occurs among its factors. In particular $S_k(G)$ is finite for a finite group G and any field k .*

Proof. A simple module S is isomorphic to R/I for a maximal left ideal I : the map $R \rightarrow S$ sending r to rs is onto for any nonzero $s \in S$. Since R has finite length, so does I . Append R to a composition series of I . The resulting series of R has final factor $R/I \simeq S$. The Jordan–Hölder theorem identifies this factor with a factor of any prescribed composition series from 0 to R .

The finite index set of the factors therefore maps onto all simple module classes. For $R = kG$, finite dimension over k makes the regular module both Noetherian and Artinian, so it has a full finite composition series. $\square$

Implementation.

`ModularRep.SimpleModuleClassFinite` proves `simpleClass_appears_in_compositionSeries` using a series whose last factor is the quotient by a maximal left ideal. `finite_simpleModuleClass_of_compositionSeries` applies finiteness to the resulting surjection. The result `finiteSimpleModuleClassGroupAlgebra` uses the chain conditions that follow from finite dimension of kG .

Proposition 5.9, together with Theorem 5.7, proves that $K_0(kG)$ is free of finite rank over $\mathbb{Z}$, for every field k .

5.6 The Grothendieck group of all modules

Proposition 5.10 (Eilenberg swindle). *For a ring R , the exact Grothendieck group of the category of all left R -modules in the chosen universe is zero.*

Proof. For any R -module M , let $T = \prod_{n \geq 0} M$. The shift isomorphism

$$M \oplus T \longrightarrow T, \quad (x, f) \longmapsto (x, f(0), f(1), \dots)$$

is R -linear. The split short exact sequence $0 \rightarrow M \rightarrow M \oplus T \rightarrow T \rightarrow 0$ gives $[M \oplus T] = [M] + [T]$. The isomorphism gives $[M \oplus T] = [T]$, so cancellation yields $[M] = 0$. Since object classes generate the group, every element is zero. $\square$

Implementation.

`ModularRep.ModuleCategoryKZeroTrivial` uses the countable product $\text{Nat} \rightarrow M$. Its results are `classOf_eq_zero` and `kZero_subsingleton`. The homomorphism of composition factors `fdRepJordanHolderKZeroHom` takes values in `FreeClassGroup (ModuleCat k[G])`, the free abelian group on isomorphism classes of all kG -modules. Imposing all short exact relations would give `KZero (ModuleCat k[G])`, which is zero by this argument.

For a nonzero kG -module M , the product $\prod_{n \geq 0} M$ is not finite dimensional over k , so the argument does not apply in $\text{FDRep}_k(G)$. This is why the restriction to finite dimensional representations is necessary in Theorem 5.7.

5.7 The Brauer character homomorphism

Now let ℓ be prime, let k be algebraically closed of characteristic ℓ , and let K be a field of characteristic zero. Fix a finite root correspondence ι for G as in Chapter 4. Let $\text{CF}_{\ell'}(G, K)$ be the additive group of K -valued class functions on the ℓ -regular elements of G .

Proposition 5.11 (Injectivity of the Brauer character homomorphism). *Taking Brauer characters induces an injective homomorphism*

$$\text{br}_{\ell} : K_0(kG) \longrightarrow \text{CF}_{\ell'}(G, K), \quad [V] \longmapsto \varphi_V.$$

Proof. Isomorphism invariance and short exact additivity of Brauer characters allow the universal property to define the homomorphism. Under the basis of Theorem 5.7, it sends e_S to the irreducible Brauer character φ_S . The functions φ_S are pairwise distinct by Proposition 4.10, and linearly independent over K by Theorem 4.11. Linear independence forces each coefficient in an integer relation to vanish in K . Since $\text{char } K = 0$, the map $\mathbb{Z} \rightarrow K$ is injective, so the coefficients vanish in $\mathbb{Z}$. $\square$

Implementation.

`ModularRep.BrauerCharacterKZero` constructs `FDRepSimpleClassKZero.brauerCharacterKZeroHom` using exact additivity. The theorem `brauerCharacterKZeroHom_injective` uses the basis of simple module classes and `irreducibleBrauerCharacterLinearIndependence_of_rootEmbedding`. The proof restricts scalars from K to $\mathbb{Z}$ and uses the injectivity of the corresponding map taking a finite integer linear combination of characters.

Thus representations with equal Brauer characters have the same composition multiplicities. In Chapter 6, character compatibility then identifies the Grothendieck classes of different lattice reductions and proves independence of the chosen lattice.

Chapter 6

Stable lattices and reduction

A stable lattice in an ordinary representation gives a modular representation by reduction modulo the maximal ideal. Different lattices may have nonisomorphic reductions, but the multiplicities of their simple composition factors agree. The decomposition homomorphism assigns this common Grothendieck class to the ordinary representation.

The construction below identifies tensor reduction with reduction modulo the maximal ideal. Lattice independence and the decomposition homomorphism then follow from the ordinary and Brauer character identity in Hypothesis 6.5.

Throughout the chapter, G is a finite group and ℓ is prime. The coefficient fields for ordinary and modular representations are K and k , respectively. The integral coefficient ring is denoted by $\mathcal{O}$. Each section states the coefficient hypotheses it uses.

6.1 Integral coefficients and the residue map

Definition 6.1. A complete ℓ -modular system $(K, \mathcal{O}, k)$ consists of a complete discrete valuation ring $\mathcal{O}$, its fraction field K of characteristic zero, a field k of characteristic ℓ , and a specified isomorphism

$$\mathcal{O}/\mathfrak{m} \simeq k,$$

where $\mathfrak{m}$ is the maximal ideal of $\mathcal{O}$. Completeness is with respect to the $\mathfrak{m}$ -adic topology. The residue map $\pi : \mathcal{O} \rightarrow k$ is the composite of the quotient map with this isomorphism.

The map π is onto and has kernel $\mathfrak{m}$. Its choice specifies the $\mathcal{O}$ -algebra structure on k used in tensor reduction. Splitting hypotheses for G and a correspondence between roots of unity in k and K are additional conditions. Algebraic closure of the fraction field K is not required.

Implementation.

The definition in `ModularRep.ModularSystem` includes the prime, the complete discrete valuation ring, its fraction field, the two characteristics, and the residue field isomorphism `residueEquiv`. The definitions `ModularSystem.residue` and `ModularSystem.residueAlgebra` give π and the induced $\mathcal{O}$ -algebra structure on k . The theorems `residue_surjective` and `ker_residue` prove the two properties of π used below. All reductions associated with this modular system use `residueAlgebra`.

6.2 Constructing a stable lattice

For this section it is enough that $\mathcal{O}$ is a commutative ring, K is a field with an $\mathcal{O}$ -algebra structure, and V is a finite dimensional K -space, with its compatible $\mathcal{O}$ -module structure. Let $\rho : G \rightarrow \mathrm{GL}_K(V)$ be a representation.

Definition 6.2. An $\mathcal{O}$ -lattice in V is a finitely generated $\mathcal{O}$ -submodule $L \leq V$ whose K -linear span is V . It is stable if $\rho(g)L \subseteq L$ for every $g \in G$.

When $\mathcal{O}$ is a discrete valuation ring with fraction field K , this agrees with the usual notion of a full lattice. Only finite generation and full span are needed for the construction below. For the usual construction over a discrete valuation ring, see, for example, [8, §15.2].

Proposition 6.3 (Stable closure of a lattice). *If L_0 is an $\mathcal{O}$ -lattice in V , then*

$$L = \sum_{g \in G} \rho(g)L_0$$

is a stable $\mathcal{O}$ -lattice containing L_0 . In particular every finite dimensional K -representation of a finite group has a stable $\mathcal{O}$ -lattice.

Proof. Each image $\rho(g)L_0$ is finitely generated over $\mathcal{O}$. There are only finitely many such submodules, so their sum is finitely generated. The term $g = 1$ shows $L_0 \subseteq L$, which implies that the K -span of L is all of V . For $h \in G$, $\rho(h)\rho(g)L_0 = \rho(hg)L_0$ is another summand. Thus $\rho(h)$ preserves L .

For existence, choose a finite K -basis $b_1, \dots, b_n$ of V and take $L_0 = \sum_i \mathcal{O}b_i$. This submodule is finitely generated and spans V over K , so the construction applies. $\square$

Implementation.

`ModularRep.StableLattice` defines `StableLattice` as a stable subrepresentation satisfying Mathlib's `IsLattice` predicate. The orbit sum is `StableLattice.orbitSpan`, represented by the supremum of the image submodules. Its containment, stability, finite generation, and full span are proved in `le_orbitSpan`, `orbitSpan_stable`, `orbitSpan_fg`, and `orbitSpan_isLattice`. The resulting lattice is `stableClosure`, and the basis construction is `ofBasis`. The construction of `ofBasis` also applies when K is a commutative ring and a finite basis is given. It requires neither a discrete valuation nor completeness.

The integral representation on L is obtained by restricting $\rho(g)$ to L for each g . Once an $\mathcal{O}$ -algebra structure on k is specified, define its reduction by

$$\bar{L} = k \otimes_{\mathcal{O}} L, \quad g(a \otimes x) = a \otimes \rho(g)x.$$

If $x_1, \dots, x_t$ generate L over $\mathcal{O}$, the tensors $1 \otimes x_1, \dots, 1 \otimes x_t$ generate $\bar{L}$ over k . Indeed expansion in the generators and the balancing relation move all $\mathcal{O}$ -coefficients to the first tensor factor. Hence the reduction is finite dimensional whenever k is a field.

Implementation.

The declarations `StableLattice.integralRepresentation` and `StableLattice.reduction` construct these actions. `reduction_apply_tmul` gives the action on pure tensors, and `reduction_moduleFinite` proves finite generation by the tensors of an arbitrary finite generating set. In `ModularRep.StableLatticeKZero`, `chosenStableLattice` uses a finite basis of an `FRep` object, and `stableLatticeReduction` is its finite dimensional reduction.

6.3 Tensor reduction as an equivariant quotient

The quotient description of reduction is valid under weaker hypotheses than those of a modular system. Let $\pi : \mathcal{O} \rightarrow k$ be a surjective homomorphism of commutative rings, and write $I = \ker \pi$. For an $\mathcal{O}$ -module M , the notation IM means the submodule generated by the products rx with $r \in I$ and $x \in M$. The quotient M/IM is a k -module through $\mathcal{O}/I \simeq k$. Write $\bar{x} = x + IM$ for the class of $x \in M$.

Proposition 6.4 (Reduction modulo an ideal). *There is a k -linear isomorphism*

$$\Phi : k \otimes_{\mathcal{O}} M \xrightarrow{\sim} M/IM, \quad a \otimes x \mapsto a\bar{x}, \quad \Phi^{-1}(\bar{x}) = 1 \otimes x.$$

If M carries an $\mathcal{O}$ -linear action of G , then IM is stable, the action descends by $g\bar{x} = \overline{gx}$, and Φ intertwines the tensor and quotient actions.

Proof. Surjectivity gives the isomorphism $\mathcal{O}/I \simeq k$. The standard tensor quotient isomorphism identifies $(\mathcal{O}/I) \otimes_{\mathcal{O}} M$ with M/IM . Using $\mathcal{O}/I \simeq k$ in the first tensor factor gives Φ with the stated formula. The quotient relation makes $\bar{x} \mapsto 1 \otimes x$ well defined: if $r \in I$, then $1 \otimes rx = \pi(r) \otimes x = 0$.

Every group operator is $\mathcal{O}$ -linear, so it sends rx to $r(gx)$ and preserves IM . This defines the quotient action directly. For a pure tensor,

$$\Phi(g(a \otimes x)) = a\overline{gx} = g(a\bar{x}) = g\Phi(a \otimes x).$$

Additivity extends the identity to the whole tensor product. □

Implementation.

`ModularRep.ReductionModulo` defines IM as `ReductionModulo.kernelSMul`. Its `tensorQuotientEquiv` combines `Mathlib's TensorProduct.quotTensorEquivQuotSMul` with its first isomorphism theorem for algebras. The local construction transports the quotient scalar action and uses surjectivity to obtain the k -linear equivalence. The versions for an explicit ring map and a specified ideal are `tensorQuotientEquivOfSurjectiveRingHom` and `tensorQuotientEquivOfKernelEq`.

`ModularRep.ReductionModuloEquivariance` first constructs `ReductionModulo.quotientRepresentation` by descending the integral action. The intertwining identity is `tensorQuotientEquiv_isIntertwining`, and the resulting equivalence of representations is `tensorQuotientRepresentationEquiv`.

For a modular system, $I = \mathfrak{m}$ and the proposition gives $\bar{L} \simeq L/\mathfrak{m}L$ as kG -modules. This is `stableLatticeReductionQuotientEquiv`. Its underlying linear equivalence agrees with `stableLatticeReductionCarrierQuotientEquiv`. For a fixed stable lattice, this isomorphism does not require G to be finite.

6.4 Character compatibility and lattice independence

Fix a complete ℓ -modular system. For the character construction in this section, additionally assume that k is algebraically closed and choose a finite root correspondence ι for G . The field K has characteristic zero and is the fraction field of $\mathcal{O}$. Write

$$\mathrm{br}_{\ell} : K_0(kG) \longrightarrow \mathrm{CF}_{\ell'}(G, K), \quad c : K_0(KG) \longrightarrow \mathrm{CF}_{\ell'}(G, K)$$

for the Brauer character homomorphism and the homomorphism taking an ordinary representation class to its trace character restricted to ℓ -regular elements. The first is injective by Proposition 5.11. The second is defined by ordinary trace additivity and the universal property of the Grothendieck group.

Hypothesis 6.5 (Reduction compatibility). Write $G_{\ell'}$ for the set of ℓ -regular elements. For every finite dimensional K -representation V , with ordinary trace character χ_V , and every stable $\mathcal{O}$ -lattice L in V , the fixed root correspondence satisfies

$$\varphi_{k \otimes \mathcal{O} L} = \chi_V|_{G_{\ell'}}.$$

The reduction identity with compatible roots is standard [6, Corollary 2.9]. Here the residue map and the root correspondence are specified separately. Their compatibility is required: the identity is assumed for every stable lattice with these same choices.

Implementation.

This hypothesis is `StableReductionBrauerCharacterCompatibility`. It is an additional condition on `ModularSystem` and `PrimeRegularRootEmbedding`.

Lattice independence and the decomposition homomorphism are treated in [8, §15.2, Theorem 32 and the following discussion]. That account proves lattice independence by comparing lattices directly. It also notes that completeness is unnecessary for that argument. Under the coefficient hypotheses and character identity stated above, lattice independence also follows from injectivity of the Brauer character homomorphism, as in the proof below.

Theorem 6.6 (Decomposition homomorphism). *Under Hypothesis 6.5, there is a unique homomorphism of abelian groups*

$$d : K_0(KG) \longrightarrow K_0(kG) \quad \text{such that} \quad \text{br}_\ell \circ d = c.$$

For every stable lattice L in V , it satisfies $d([V]) = [k \otimes_{\mathcal{O}} L]$. Consequently two stable lattices in the same ordinary representation have equal modular Grothendieck classes.

Proof. Choose one stable lattice for every ordinary representation, using Proposition 6.3, and assign its reduction class to the representation. If L_1, L_2 lie in the same V , compatibility gives

$$\text{br}_\ell([\overline{L}_1]) = c([V]) = \text{br}_\ell([\overline{L}_2]).$$

Injectivity implies $[\overline{L}_1] = [\overline{L}_2]$. The same argument using invariance of c under isomorphism compares choices for isomorphic ordinary representations.

For a short exact sequence with terms U, V, W , let a_U, a_V, a_W be the chosen reduction classes. Ordinary trace additivity gives

$$\text{br}_\ell(a_V) = c([V]) = c([U]) + c([W]) = \text{br}_\ell(a_U + a_W).$$

Injectivity gives $a_V = a_U + a_W$. The universal property therefore defines d . On every object class, compatibility gives $\text{br}_\ell d = c$, so the two homomorphisms agree everywhere. Comparing $d([V])$ with any other stable lattice reduction proves the asserted formula for every L . Finally, if d' has the same character identity, then $\text{br}_\ell(d'(x)) = \text{br}_\ell(d(x))$ for every x . Injectivity gives $d' = d$. $\square$

Implementation.

`ModularRep.ExactCharacterBridge` proves the same assertion for homomorphisms from the two Grothendieck groups to an abelian group A , provided the modular homomorphism is injective and the character identity holds for every stable lattice. The resulting definitions are `ExactCharacterBridge.toExactDecompositionMapData` and `ExactCharacterBridge.decompositionMap`.

`ModularRep.BrauerDecompositionMap` specialises this argument to ordinary and Brauer characters. Its `exactCharacterBridgeOfStableReduction` uses

`brauerCharacterKZeroHom_injective` for injectivity on $K_0(kG)$. The resulting homomorphism is `decompositionMapOfStableReduction`, its character identity is `decompositionMapOfStableReduction_characterSquare`, and its uniqueness is `existsUnique_decompositionMapOfStableReduction`. The declarations `decompositionMap_classOf_eq_stableLatticeReduction` and `stableLatticeReductionClass_eq` give the two lattice statements.

By Theorem 5.7, equality of these classes is equivalent to equality of their composition multiplicities.

Example 6.7 (Decomposition numbers). For a simple kG -module S , consider the coefficient of $[S]$ in $d([V])$ with respect to the basis of simple module classes. If L is any stable lattice in V , this coefficient is $m_S(\overline{L})$ and is a nonnegative integer. For a short exact sequence $0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$, it is the sum of the coefficients for U and W . For irreducible V , these coefficients are the decomposition numbers. This definition makes sense over the coefficient fields fixed here. To identify them with the usual character decomposition numbers for G , one also chooses K to be a splitting field for G .

6.5 Twists and tensor products

For $\alpha \in \text{Aut}(G)$, define the twist of V by precomposition: $\rho^\alpha(g) = \rho(\alpha(g))$. The induced operation on exact K_0 is denoted by T_α . It comes from an exact autoequivalence, and the same notation is used over K and k .

Proposition 6.8 (Compatibility with automorphisms). *Under the hypotheses of Theorem 6.6,*

$$d(T_\alpha x) = T_\alpha d(x) \quad (\alpha \in \text{Aut}(G), x \in K_0(KG)).$$

Proof. Both character homomorphisms commute with precomposition by α . Applying br_ℓ to the two sides therefore gives the same function, namely $c(x) \circ \alpha$ on G_ℓ . Injectivity proves the equality in $K_0(kG)$. $\square$

Implementation.

`ModularRep.KZeroTwist` defines `FDRep.twistEquivalence` by applying Mathlib's `Action.resEquiv` to finite dimensional representations. It constructs the induced map `twistKZero` and proves `ordinaryCharacterKZero_twist` and `brauerCharacterKZeroHom_twist`. The consequence above is `decompositionMapOfStableReduction_twist`. The convention is a right action: with automorphism multiplication given by composition, `twistKZero_comp` states $T_\alpha \circ T_\beta = T_{\beta\alpha}$.

For a linear character $\lambda : G \rightarrow k^\times$, write $L_\lambda V$ for the representation on the same vector space with action $g \cdot v = \lambda(g)\rho(g)v$. This is tensoring by the corresponding one dimensional representation. Tensoring by λ^{-1} is its inverse, so it is an exact autoequivalence. Use L_λ also for the induced operation on K_0 .

Twisting and tensoring satisfy

$$T_\alpha L_\lambda = L_{\lambda \circ \alpha} T_\alpha.$$

The two composite exact functors are naturally isomorphic: the identity on the representation space intertwines their equal actions. This argument applies over either coefficient field and requires neither a modular system nor a formula for Brauer characters of tensor products. The natural isomorphism is `ModularRep.FDRep.linearCharacterTwistTwistIso`, and its consequence on K_0 is `ModularRep.twistKZero_linearCharacterTwistKZero`, both in `ModularRep.KZeroLinearCharacterSemidirect`.

For a modular linear character λ , write $\widehat{\lambda}(g)$ for the lift of $\lambda(g)$ through ι , for $g \in G_{\ell'}$. Let $\lambda_K : G \rightarrow K^\times$ and $\lambda_k : G \rightarrow k^\times$ be linear characters, and assume that $\lambda_K|_{G_{\ell'}} = \widehat{\lambda_k}$. By Theorem 4.16,

$$\varphi_{L_\lambda V} = \widehat{\lambda} \varphi_V \quad (V \in \text{FDRep}_k(G), \lambda : G \rightarrow k^\times)$$

on $G_{\ell'}$, with pointwise multiplication on the right. For the general tensor product formula, see [6, proof of Theorem 2.23].

Proposition 6.9 (Compatibility with linear characters). *If d is the homomorphism of Theorem 6.6 and λ_K and λ_k satisfy the compatibility condition above, then, for every $x \in K_0(KG)$,*

$$d(L_{\lambda_K} x) = L_{\lambda_k} d(x).$$

Consequently the combined operation also satisfies $d(L_{\lambda_K} T_\alpha x) = L_{\lambda_k} T_\alpha d(x)$.

Proof. Ordinary trace on $L_{\lambda_K} V$ is the product of λ_K with the trace on V , since scalar multiplication by $\lambda_K(g)$ scales the trace by the same factor. Theorem 4.16 gives the analogous statement over k . Compatibility of the two linear characters therefore identifies the images under br_ι of $d(L_{\lambda_K} x)$ and $L_{\lambda_k} d(x)$. Injectivity gives the first identity. Combining it with Proposition 6.8 gives the second. $\square$

Implementation.

`ModularRep.KZeroLinearCharacterTensor` constructs the exact autoequivalence and `linearCharacterTwistKZero`, and proves the ordinary trace formula. In `ModularRep.KZeroLinearCharacterDecomposition`, `LinearCharacterReductionCompatible` states the equality $\lambda_K|_{G_{\ell'}} = \widehat{\lambda_k}$. The conclusions are `decompositionMapOfStableReduction_linearCharacterTwist` and `decompositionMapOfStableReduction_linearCharacterTwist_twist`. Both the compatibility of the linear characters and `BrauerLinearTensorProductFormula` are hypotheses of these results.

6.6 Specifying a decomposition homomorphism

An assignment $[V] \mapsto a_V \in K_0(kG)$ that is additive on short exact sequences induces a homomorphism $K_0(KG) \rightarrow K_0(kG)$. If $a_V = [\overline{L}]$ for every stable lattice L in V , this is the decomposition homomorphism. Theorem 6.6 proves these properties from character compatibility.

Implementation.

`ExactDecompositionMapData` describes an assignment on ordinary isomorphism classes that is additive on short exact sequences. Its `toAddMonoidHom` is the homomorphism given by the universal property. The condition `RealisesStableLatticeReduction` says that every stable lattice has the assigned reduction class. For `decompositionMapOfStableReduction`, this condition is proved by `exactDecompositionMapDataOfStableReduction_realises`.

`DecompositionMapInterface` describes a homomorphism between specified ordinary and modular class groups that agrees with reduction for every stable lattice. Equality of the classes from two lattices then follows by expressing both as the image of the same ordinary class.

Chapter 7

Central characters and blocks

A central idempotent gives a direct product decomposition of a ring. For a group algebra, primitive central idempotents define the blocks. Central subgroup idempotents relate the scalar action on a simple module to its block. The Brauer homomorphism and block induction relate blocks of different groups. They give Brauer's first main theorem under the stated facts about defect groups. We also bound the group of linear characters stabilising a block.

Throughout, G is a finite group. Unless a stronger assumption is stated, k is a field. In arguments of positive characteristic, ℓ denotes a prime and $\text{char } k = \ell$. For $z = \sum_{g \in G} z_g g \in kG$, the symbol z_g denotes its coefficient at g . A block means a primitive central idempotent of kG . When blocks are indexed by a finite set $\mathcal{B}(G)$, their idempotents are denoted by b_B , for $B \in \mathcal{B}(G)$. A complete primitive decomposition and the central characters of its blocks will be assumed where they are needed.

7.1 Central idempotents and simple modules

The initial arguments apply to an arbitrary unital ring R .

Definition 7.1. A complete family of orthogonal central idempotents is a finite family $(e_i)_{i \in I}$ in R such that each e_i is central, $e_i^2 = e_i$, $e_i e_j = 0$ for $i \neq j$, and $\sum_i e_i = 1$. A nonzero central idempotent b is *primitive* if every central idempotent c with $cb = c$ is either 0 or b . A module *belongs to the block* b if b acts as the identity.

Proposition 7.2 (Central idempotents on simple modules). *Let $(e_i)_{i \in I}$ be a complete family of orthogonal central idempotents in R . Every simple R -module V has a unique index i such that e_i acts as the identity. Every other e_j acts as zero. Every primitive central idempotent b likewise has a unique index i such that*

$$be_i = b, \quad be_j = 0 \quad (j \neq i).$$

If b acts as the identity on V , the two indices coincide.

Proof. Centrality makes $v \mapsto e_i v$ an R -linear endomorphism. On a simple module it is either zero or bijective. In the latter case its idempotence implies that it is the identity: injectivity applied to $e_i(e_i v) = e_i v$ gives $e_i v = v$. If every e_i acted as zero, then $1 = \sum_i e_i$ would act as zero on the nonzero module V . Thus at least one acts as the identity. If e_i does, then $e_j v = e_j e_i v = 0$ for $j \neq i$, proving uniqueness.

For the assertion about b , each be_i is a central idempotent with $(be_i)b = be_i$. Primitivity makes it either zero or b . The equation $\sum_i be_i = b \neq 0$ gives a nonzero term. If $be_i = b$,

multiplication by e_j and orthogonality give $be_j = 0$ for $j \neq i$. Finally, if b acts as the identity, then $e_j v = b(e_j v) = (be_j)v$, so the two indices coincide. $\square$

Implementation.

The structure for a complete family of central idempotents extends Mathlib's `CompleteOrthogonalIdempotents`. The primitivity predicate is local, and the proof for simple modules uses Mathlib's Schur lemma. The two existence and uniqueness theorems are `ModularRep.CompleteOrthogonalCentralIdempotents.existsUnique_isSupport` and `ModularRep.IsPrimitiveCentralIdempotent.existsUnique_isSupport`. The agreement of the two indices is `ModularRep.IsPrimitiveCentralIdempotent.IsSupport.toModuleSupport`. Their modules are `ModularRep.CentralIdempotentSupport` and `ModularRep.PrimitiveCentralIdempotent`.

A *block decomposition* is a complete family of orthogonal primitive central idempotents. For the expansion of central idempotents in the group algebra case, see, for example, [6, Theorem 3.11].

Proposition 7.3 (Expansion in primitive idempotents). *Let $(b_B)_{B \in \mathcal{B}}$ be a block decomposition of R , and let e be a central idempotent. Set $S(e) = \{B \in \mathcal{B} : b_B e \neq 0\}$. Then*

$$b_B e \in \{0, b_B\}, \quad e = \sum_{B \in S(e)} b_B.$$

The set $S(e)$ is the unique subset of $\mathcal{B}$ with this sum. In particular, $e \neq 0$ if and only if $S(e)$ is nonempty.

Proof. The product $b_B e$ is a central idempotent below b_B , so primitivity gives the two alternatives. Consequently

$$e = 1e = \sum_B b_B e = \sum_{B \in S(e)} b_B.$$

Multiplication of a proposed sum by b_C gives b_C when C is in its index set and zero otherwise. Since $b_C \neq 0$, this recovers the index set from the sum and proves uniqueness. An empty sum is zero, whereas a nonempty sum has a nonzero product with any of its members. $\square$

Implementation.

`ModularRep.CentralIdempotentBlockExpansion` proves the expansion and its uniqueness in `ModularRep.CompleteOrthogonalCentralIdempotents.sum_centralIdempotentSupport_eq` and `ModularRep.CompleteOrthogonalCentralIdempotents.eq_centralIdempotentSupport_of_sum_eq`. For a given block decomposition, `ModularRep.BlockIdempotentDecomposition` defines `moduleBlock` and the equivalence `primitiveBlockEquiv` between the indexing set and the primitive central idempotents.

Example 7.4 (Using a decomposition). Given a simple representation and a block decomposition, the unique idempotent acting as the identity determines its block. An isomorphism of modules preserves this index because it preserves each equation $b_B v = v$. This is proved in `ModularRep.BlockIdempotentDecomposition.moduleBlock_eq_of_linearEquiv`.

7.2 Idempotents of central subgroups

Let $Z \leq Z(G)$ have order invertible in k . For a linear character $\nu : Z \rightarrow k^\times$, define

$$s_\nu = \sum_{z \in Z} \nu(z)^{-1} z, \quad e_\nu = |Z|^{-1} s_\nu \in kG.$$

The inverse in the coefficient ensures that e_ν acts as the identity on a module with scalar action $zv = \nu(z)v$.

Proposition 7.5 (Idempotents associated with central characters). *Each e_ν is a central idempotent, and $e_\nu e_\mu = 0$ for $\nu \neq \mu$. If k is algebraically closed, then*

$$\sum_{\nu \in \text{Hom}(Z, k^\times)} e_\nu = 1.$$

Thus the idempotents form a complete family of orthogonal central idempotents.

Proof. For a nontrivial linear character χ of a finite group, choose a with $\chi(a) \neq 1$. Translation by a permutes the group, so

$$\chi(a) \sum_z \chi(z) = \sum_z \chi(az) = \sum_z \chi(z).$$

Cancellation in the field gives $\sum_z \chi(z) = 0$. For the trivial character the sum is the group order. Reindexing the product of the weighted sums by $t = xy$ gives

$$s_\nu s_\mu = \left(\sum_{x \in Z} \nu(x)^{-1} \mu(x) \right) s_\mu.$$

This is $|Z|s_\nu$ when $\nu = \mu$ and zero otherwise. Division by $|Z|^2$ proves idempotence and orthogonality. Each group element in the sum is central in G , which proves centrality in kG .

For completeness, finite abelian duality applies because k has enough roots of unity for the exponent of Z . Its character group has order $|Z|$ and separates the elements of Z . If $z \neq 1$, evaluation at z^{-1} is therefore a nontrivial character of this dual group. The same translation argument gives

$$\sum_\nu \nu(z)^{-1} = 0 \quad (z \neq 1), \quad \sum_\nu \nu(1)^{-1} = |Z|.$$

These are exactly the coefficient equations for $\sum_\nu e_\nu = 1$. Algebraic closure and invertibility of $|Z|$ provide the required roots. $\square$

Implementation.

The weighted product is `ModularRep.linearCharacterWeightedSum_mul` in `ModularRep.LinearCharacterOrthogonality`. The completeness proof is `ModularRep.sum_centralCharacterIdempotent_eq_one` in `ModularRep.LinearCharacterCompleteness`. It uses Mathlib's finite abelian duality to count the characters and separate group elements, with the required roots of unity specified. The algebraically closed version is proved in `ModularRep.completeCentralCharacterIdempotentsOfIsAlgClosed`.

For a finite dimensional irreducible representation ρ over an algebraically closed field, Schur's lemma gives a unique linear character $\nu_\rho : Z \rightarrow k^\times$ such that $\rho(z) = \nu_\rho(z) \text{id}$. The block central character ω_B introduced below has domain $Z(kG)$, whereas ν_ρ has domain Z .

Proposition 7.6 (The central character of a block). *Assume k is algebraically closed and $|Z|$ is invertible in k . Every primitive central idempotent b lies above a unique character ν_b such that*

$$be_{\nu_b} = b, \quad be_\mu = 0 \quad (\mu \neq \nu_b).$$

If ρ is finite dimensional and irreducible, and b acts as the identity on ρ , then $\nu_\rho = \nu_b$. Hence any two such representations have the same scalar action of Z .

Proof. Apply Proposition 7.2 to the complete family of Proposition 7.5. On ρ , the action of e_μ is multiplication by

$$|Z|^{-1} \sum_{z \in Z} \mu(z)^{-1} \nu_\rho(z),$$

which is one for $\mu = \nu_\rho$ and zero otherwise. Since b acts as the identity on ρ , Proposition 7.2 gives $\nu_b = \nu_\rho$. $\square$

Implementation.

`ModularRep.IsPrimitiveCentralIdempotent.existsUnique_centralCharacterSector` and `Representation.primitiveCentralIdempotentSector_eq_centralCharacter` are in `ModularRep.CentralCharacterBlockSector`. The scalar calculation is in `ModularRep.CentralCharacterAction`. The equality of central characters assumes an irreducible representation belonging to the block.

Example 7.7 (Automorphisms and central characters). Assume k is algebraically closed. A ring isomorphism preserves primitivity: pull back a central idempotent below the image of b , apply primitivity of b , then map forward. For a group automorphism α , transport the basis of the group algebra by α^{-1} . If the order of $Z(G)$ is invertible in k , coefficient comparison gives

$$e_\nu \longmapsto e_{\nu \circ \alpha|_{Z(G)}}.$$

Thus a block lying above ν is sent to a block lying above $\nu \circ \alpha|_{Z(G)}$. The corresponding theorem is `ModularRep.IsPrimitiveCentralIdempotent.centralCharacterSector_mapDomainRingEquiv_center`. This example concerns the whole centre under the stated invertibility hypothesis, as in `ModularRep.PrimitiveBlockAutomorphism`.

7.3 Block central characters and induction

Assume in this section that k is algebraically closed, and fix a block decomposition. The following standard description of block central characters is used as an explicit input to the Lean argument. In positive characteristic it is given by [6, Theorem 3.11].

Hypothesis 7.8 (Central characters of blocks). For each $B \in \mathcal{B}(G)$ there is a k -algebra homomorphism $\omega_B : Z(kG) \rightarrow k$ satisfying

$$\omega_B(b_C) = \begin{cases} 1 & B = C, \\ 0 & B \neq C. \end{cases}$$

Every k -algebra homomorphism $Z(kG) \rightarrow k$ is one of the ω_B .

The evaluation equations imply that distinct blocks have distinct central characters: evaluate an equality $\omega_B = \omega_C$ at b_B . Thus $B \mapsto \omega_B$ is a bijection between the blocks and the k -algebra homomorphisms $Z(kG) \rightarrow k$. The central character ω_B does not generally describe multiplication on the block algebra: $zb_B = \omega_B(z)b_B$ need not hold for $z \in Z(kG)$.

Implementation.

The hypotheses are expressed by `ModularRep.BlockCentralCharacterCatalogue`. The bijection is `ModularRep.BlockCentralCharacterCatalogue.centralCharacterEquiv`.

For $H \leq G$, coefficient restriction is the linear map

$$r_H : kG \longrightarrow kH, \quad r_H \left(\sum_{g \in G} z_g g \right) = \sum_{h \in H} z_h h.$$

It carries $Z(kG)$ into $Z(kH)$: the coefficients of a central element are constant on G -conjugacy classes, so the restricted coefficients are constant on H -conjugacy classes. Thus $r_H(z)$ commutes with every basis element of kH . Also $r_H(1) = 1$. Multiplicativity does not hold for a general subgroup. Even for an abelian group and $H = 1$, an element $g \neq 1$ has $r_H(g) = r_H(g^{-1}) = 0$ but $r_H(gg^{-1}) = 1$.

Definition 7.9. We use the definition of block induction by central characters (see, for example, [6, p. 87]). Given the block central characters for H and G , block induction from a block $b \in \mathcal{B}(H)$ is *defined* when the linear function $\omega_b \circ r_H : Z(kG) \rightarrow k$ is multiplicative. A block $B \in \mathcal{B}(G)$ is *induced from* b if this condition holds and $\omega_B = \omega_b \circ r_H$.

Proposition 7.10 (Existence and uniqueness of an induced block). *If block induction from b is defined, there is exactly one induced block B .*

Proof. The composite is k -linear, multiplicative and sends one to one. It is therefore a k -algebra homomorphism. By Hypothesis 7.8, it equals ω_B for some B . Uniqueness follows from injectivity of $B \mapsto \omega_B$, proved by evaluation at block idempotents. $\square$

Implementation.

`ModularRep.GroupAlgebraCentralFunctions` uses Mathlib's group algebra and its centre. It proves centrality of coefficient restriction and constructs `centerCoeffRestrict`. `ModularRep.BlockInduction` defines `IsBlockInductionDefined`, proves `existsUnique_blockInducesTo`, and defines `inducedBlock`. The induced block is defined from multiplicativity and the given block central characters. The external source interfaces in `ModularRep.SpathNavarroSourceAdapter` use an algebraically closed field k of characteristic ℓ that is algebraic over $\mathbb{F}_\ell$. This additional hypothesis is recorded by `ModularRep.SpathCoefficientField`.

Proposition 7.11 (Preservation on a central subgroup). *Let $Z \leq Z(G) \cap H$, with $|Z|$ invertible in k . Suppose $b \in \mathcal{B}(H)$ induces to $B \in \mathcal{B}(G)$. Let V and W be finite dimensional irreducible representations of H and G , belonging to the blocks b and B , respectively. After identifying Z with its copy in H , their scalar characters on Z agree.*

Proof. For a central idempotent $e \in kG$, primitivity gives $b_B e \in \{0, b_B\}$. Applying ω_B shows

$$\omega_B(e) = 1 \iff b_B e = b_B.$$

The reverse implication uses $\omega_B(b_B) = 1$. In the forward implication the alternative $b_B e = 0$ would give $1 = 0$.

Choose ν so that b_B lies above ν on Z . Since $Z \leq H$, e_ν is supported on H , so $r_H(e_\nu)$ is the corresponding character idempotent in kH . In particular it is still idempotent. Induction gives

$$\omega_b(r_H(e_\nu)) = \omega_B(e_\nu) = 1.$$

The same criterion in kH gives $b_b r_H(e_\nu) = b_b$. Hence b_b also lies above ν . Proposition 7.6 identifies ν with the scalar characters of V and W . $\square$

Implementation.

The criterion for the value on a central idempotent is `ModularRep.BlockCentralCharacterCatalogue.centralIdempotent_value_one_iff`. The conclusion is `Representation.centralCharacter_eq_comp_of_blockInducesTo` in `ModularRep.CentralCharacterBlockInductionRestriction`. The proof identifies Z with its copy in H and uses the resulting equality of orders. It does not require V to occur in the restriction of W .

7.4 The central Brauer map

Now assume $\text{char } k = \ell$ and let $P \leq G$ be an ℓ -subgroup. Write $C = C_G(P)$. The *central Brauer map* is coefficient restriction to C on the centre of kG :

$$\text{Br}_P : Z(kG) \longrightarrow Z(kC), \quad \text{Br}_P(z) = \sum_{c \in C} z_c c.$$

The target is the centre of kC , by the restriction argument above. Its multiplicativity follows from an orbit calculation. For this case $H = C_G(P)$, see, for example, [6, Theorem 4.9].

Lemma 7.12 (Summing over fixed points). *Let an ℓ -group Q act on a finite set X . If $f : X \rightarrow k$ is constant on each Q -orbit, then*

$$\sum_{x \in X} f(x) = \sum_{x \in X^Q} f(x).$$

Proof. Partition X into orbits. On an orbit $\mathcal{O}$, the sum equals $|\mathcal{O}|f(x)$ for any $x \in \mathcal{O}$. Orbit sizes are powers of ℓ , so an orbit of size greater than one contributes zero in characteristic ℓ . An orbit of size one is exactly a fixed point and contributes its own value. $\square$

Theorem 7.13 (The Brauer map on the centre). *For a finite group G , an ℓ -subgroup P and a field k of characteristic ℓ , the map Br_P is a k -algebra homomorphism. Algebraic closure is not required.*

Proof. Fix $c \in C$. The coefficient of zw at c is

$$(zw)_c = \sum_{a \in G} z_a w_{a^{-1}c}.$$

Let P act on G by conjugation. If $q \in P$, then $qc = cq$, so

$$(qaq^{-1})^{-1}c = q(a^{-1}c)q^{-1}.$$

Centrality of z and w makes the function $a \mapsto z_a w_{a^{-1}c}$ constant on these orbits. Its fixed points are precisely $a \in C$. Lemma 7.12 gives

$$(zw)_c = \sum_{a \in C} z_a w_{a^{-1}c} = (\text{Br}_P(z) \text{Br}_P(w))_c.$$

Here $a^{-1}c \in C$ whenever $a, c \in C$, so the final expression is the product coefficient in kC . Equality at every $c \in C$ proves multiplicativity. Linearity and preservation of one were already proved for coefficient restriction. $\square$

Implementation.

The proof uses Mathlib's theorem that orbit sizes for an ℓ -group are powers of ℓ . The resulting sum formula is `ModularRep.IsPGroup.sum_eq_sum_fixedPoints_of_invariant` in `ModularRep.PGroupInvariantSum`. Multiplicativity and the algebra homomorphism are `ModularRep.centralBrauerRestriction_mul` and `ModularRep.centralBrauerMap` in `ModularRep.GroupAlgebraCentralBrauerMap`. These declarations allow any commutative coefficient semiring of characteristic ℓ .

Proposition 7.14 (The Brauer map to an intermediate subgroup). *Suppose $C_G(P) \leq H \leq N_G(P)$. Extend the coefficients of $\text{Br}_P(z)$ by zero from $C_G(P)$ to H . The resulting map*

$$\text{Br}_P^H : Z(kG) \longrightarrow Z(kH)$$

is a k -algebra homomorphism. Moreover, $\text{Br}_P(z) \neq 0$ if and only if $\text{Br}_P^H(z) \neq 0$.

Proof. The inclusion $kC_G(P) \hookrightarrow kH$ is an injective algebra homomorphism. Every $h \in H$ normalises P , and therefore preserves $C_G(P)$ by conjugation. The coefficients z_c are constant under this conjugation because $z \in Z(kG)$. The extended element thus lies in $Z(kH)$. Multiplication and the unit are preserved by the composite. Injectivity gives the asserted equivalence of nonvanishing. $\square$

For this form of the Brauer homomorphism, see, for example, [6, Theorem 4.9]. Its image is central in kH because the restricted coefficients are invariant under H . An arbitrary element of $Z(kC_G(P))$ need not have this property.

Implementation.

The homomorphism `ModularRep.centralBrauerMapTo` is in `ModularRep.GroupAlgebraCentralBrauerMapTo`. The equivalence of nonvanishing, `ModularRep.centralBrauerMap_ne_zero_iff_mapTo_ne_zero`, is in `ModularRep.CentralBrauerIntervalSupport`. For $H = N_G(P)$, the map is named `ModularRep.normalizerCentralBrauerMap`.

Example 7.15 (A local sum of blocks). The image of a primitive central idempotent of kG under Br_P^H is a central idempotent, although it can be zero or a sum of several primitive central idempotents of kH . A block decomposition of kH and Proposition 7.3 determine its unique support. This is implemented by `ModularRep.sum_centralBrauerMapToLocalSupport_eq` and `ModularRep.centralBrauerMapToPrimitiveImage_ne_zero_iff_support_nonempty`. In particular, the Brauer homomorphism need not preserve primitivity.

7.5 Brauer's first main theorem

Brauer's first main theorem identifies the blocks of G with defect group P with the blocks of $N_G(P)$ having defect group P . See, for example, [6, Theorem 4.17]. We prove the correspondence from the central character identity and the five facts about defect groups stated below. Assume k is algebraically closed of characteristic ℓ . Fix an ℓ -subgroup P , write $N = N_G(P)$, and fix block decompositions and block central characters for N and G as in Hypothesis 7.8.

We first use the central character identity (see, for example, [6, Theorem 4.14]). For $PC_G(P) \leq H \leq N_G(P)$, it is

$$\omega_b \circ r_H = \omega_b \circ \text{Br}_P^H \quad (b \in \mathcal{B}(H)). \quad (7.1)$$

The hypothesis includes $P \leq H$, whereas the construction of Br_P^H only required $C_G(P) \leq H \leq N_G(P)$. Since the right side of (7.1) is an algebra homomorphism, block induction is defined. For $H = N$, denote the induced block map by $I : \mathcal{B}(N) \rightarrow \mathcal{B}(G)$. Thus

$$\omega_{I(b)} = \omega_b \circ \text{Br}_P^N. \quad (7.2)$$

For the following identity, see, for example, the final assertion of [6, Theorem 4.14], with $H = N_G(P)$.

Proposition 7.16 (The Brauer image as a sum of blocks). *Under (7.1), for every block B of G ,*

$$\text{Br}_P^N(b_B) = \sum_{b \in \mathcal{B}(N): I(b)=B} b_b.$$

In particular, if the image is nonzero, some block of N induces to B .

Proof. Expand the image as in Proposition 7.3. A local block b lies in its support exactly when its central character takes value one on the image. Indeed, multiplication by b_b is either zero or

b_b , and evaluation by ω_b distinguishes these alternatives. Equation (7.2) identifies this value with $\omega_{I(b)}(b_B)$. The evaluation equations for the ambient block central characters say that this is one exactly when $I(b) = B$. $\square$

Implementation.

`ModularRep.Navarro414IntervalCentralCharacterSource` expresses (7.1), from which the induced block is defined. The equivalence between occurrence in this sum and induction to B is `ModularRep.mem_navarro417PhaseANormalizerSupport_iff_inductionMap_eq` in `ModularRep.Navarro417PhaseASupport`.

For the Brauer map characterisation of defect groups, see, for example, [6, Theorem 4.11]. For a block B of a finite group X , let $\mathcal{D}_X(B)$ be the set of ℓ -subgroups $D \leq X$ such that

$$\mathrm{Br}_Q(b_B) \neq 0 \iff Q \text{ is conjugate to a subgroup of } D \text{ for every } \ell\text{-subgroup } Q \leq X. \quad (7.3)$$

In the following argument, we use this criterion to define $\mathcal{D}_X(B)$. Its members are the defect groups of B . Any two members are conjugate: the criterion makes each conjugate to a subgroup of the other, and equality of their finite orders then gives conjugacy. Existence will be used explicitly below.

Hypothesis 7.17 (Standard facts for the defect correspondence). We use the following five standard facts about the fixed block decompositions and central characters.

1. If $P \in \mathcal{D}_N(b)$ and an N -conjugacy class $\mathcal{K}$ has nonzero coefficient in b_b , then $\mathcal{K} = \mathcal{L} \cap C_G(P)$ for some G -conjugacy class $\mathcal{L}$, with the intersection viewed inside N .
2. The set $\mathcal{D}_N(b)$ is nonempty for every block b of N .
3. If $D \in \mathcal{D}_N(b)$, then $O_\ell(N) \leq D$.
4. If b induces to B and $D \in \mathcal{D}_N(b)$, then there is $E \in \mathcal{D}_G(B)$ containing D , viewed as a subgroup of G .
5. If $P \in \mathcal{D}_N(b)$, then there is $E \in \mathcal{D}_G(I(b))$ with $E \leq P$.

For the first condition, Lemma 4.15 restricts the nonzero class coefficients to classes with defect group P , and Lemma 4.16 writes each such class as the required intersection [6, pp. 88–89]. The second condition follows from existence of defect groups and their Brauer map characterisation [6, Theorems 4.3 and 4.11]. The third and fourth conditions follow from [6, Theorem 4.8 and Lemma 4.13], respectively.

The fifth condition is the containment proved in the first paragraph of the proof of [6, Theorem 4.17]. That argument chooses a defect class of b , uses Corollary 4.5 and Lemma 4.16 to lift it to an ambient class with defect group P , and applies Theorems 4.14 and 4.4 to obtain a defect group of $I(b)$ contained in P . We use this containment, together with the other four conditions and (7.1), to prove the correspondence.

Proposition 7.18 (Lifting a local block idempotent). *Assume the first condition of Hypothesis 7.17 and (7.1). If $P \in \mathcal{D}_N(b)$, then b_b is in the image of Br_P^N . If $I(b) = I(c)$ for any local block c , then $b = c$. Consequently,*

$$\mathrm{Br}_P^N(b_{I(b)}) = b_b.$$

Proof. For an N -class $\mathcal{K}$, write $\mathcal{K}^+ = \sum_{x \in \mathcal{K}} x$. Centrality gives a unique class expansion

$$b_b = \sum_{\mathcal{K}} a_{\mathcal{K}} \mathcal{K}^+.$$

For each nonzero coefficient choose an ambient class $\mathcal{L}_\mathcal{K}$ with $\mathcal{L}_\mathcal{K} \cap C_G(P) = \mathcal{K}$. Coefficient restriction gives $\text{Br}_P^N(\mathcal{L}_\mathcal{K}^+) = \mathcal{K}^+$. Thus the finite sum

$$z = \sum_{a_\mathcal{K} \neq 0} a_\mathcal{K} \mathcal{L}_\mathcal{K}^+ \in Z(kG) \quad \text{satisfies} \quad \text{Br}_P^N(z) = b_b.$$

If $I(b) = I(c)$, equation (7.2) gives equal composites $\omega_b \text{Br}_P^N = \omega_c \text{Br}_P^N$. Evaluation at z gives $\omega_b(b_b) = \omega_c(b_b)$. The left side is one, and the right side is zero unless $b = c$. This proves injectivity even when only the first block is known to have defect P . In the sum of Proposition 7.16, the only block inducing to $I(b)$ is therefore b . This gives the final equation. $\square$

Implementation.

The element z is constructed in `ModularRep.Navarro415416ClassSumAdapter`, in the theorem `ModularRep.Navarro415416RawClassSource.localBlockIdempotent_mem_normalizerCentralBrauerMap_range`. Injectivity and the equality of block idempotents are `ModularRep.navarro417PhaseAInductionMap_eq_imp_eq_of_localP_left` and `ModularRep.normalizerCentralBrauerMap_selectedBlockIdempotent_eq` in `ModularRep.Navarro417RestrictedInjectivity`. The class intersection property and the central character equality suffice for all three conclusions.

Theorem 7.19 (Blocks with a prescribed defect group). *Assume (7.1) and Hypothesis 7.17. Write $\mathcal{B}(X \mid P) = \{B \in \mathcal{B}(X) : P \in \mathcal{D}_X(B)\}$, where $X = G$ or $N_G(P)$ and P is viewed as a subgroup of X . Then block induction is a bijection*

$$I : \mathcal{B}(N_G(P) \mid P) \longrightarrow \mathcal{B}(G \mid P).$$

Proof. Let $b \in \mathcal{B}(N \mid P)$. The fourth condition gives $E_+ \in \mathcal{D}_G(I(b))$ with $P \leq E_+$. The fifth gives $E_- \in \mathcal{D}_G(I(b))$ with $E_- \leq P$. They are conjugate by (7.3), so

$$|P| \leq |E_+| = |E_-| \leq |P|.$$

The inclusions are therefore equalities. In particular, $P \in \mathcal{D}_G(I(b))$.

Conversely, suppose $I(b) = B$ and $P \in \mathcal{D}_G(B)$. By the second condition, choose $D \in \mathcal{D}_N(b)$. Since P is a normal ℓ -subgroup of N , the third condition gives

$$P \leq O_\ell(N) \leq D.$$

The fourth condition gives $E \in \mathcal{D}_G(B)$ with $D \leq E$ inside G . The defect groups E and P are conjugate. Hence $|P| \leq |D| \leq |E| = |P|$, and the local inclusion is an equality. Thus $P \in \mathcal{D}_N(b)$ for every local block inducing to B .

Finally, if $P \in \mathcal{D}_G(B)$, then $\text{Br}_P(b_B) \neq 0$ by (7.3). Proposition 7.14 preserves this nonvanishing in kN , and Proposition 7.16 supplies a local block over B . The preceding paragraph puts that block in $\mathcal{B}(N \mid P)$, proving surjectivity. Injectivity follows from Proposition 7.18. $\square$

Implementation.

`ModularRep.Navarro417ExactDefectAdapter` proves the two containment arguments and the bijection as `navarro417PhaseAInductionMap_has_ambientP`, `navarro417LocalP_of_blockInducesTo_ambientP`, and `navarro417ExactDefectMap_bijective`, all in namespace `ModularRep`. The resulting equivalence is `ModularRep.navarro417ExactDefectEquiv`. The construction takes the five conditions above and (7.1), for the given block decompositions and central characters. The third condition, containment of the normal ℓ -core in every defect group, is proved by `navarro408PCoreDefectSource` in `ModularRep.PaperProofs.SporadicFi24P3Definition44NamedCarrierPCoreDefectContainment`. The remaining source conditions used by this construction are stated separately.

7.6 Linear characters stabilising a block

Let $N \trianglelefteq G$ and assume again that k is algebraically closed of characteristic ℓ . Write $\pi : G \rightarrow G/NZ(G)$ for the quotient homomorphism. A linear character $\lambda : G \rightarrow k^\times$ twists a representation by

$$(\lambda \otimes \rho)(g) = \lambda(g)\rho(g).$$

If a central subgroup Z acts on ρ by a scalar character ν_ρ , it acts on $\lambda \otimes \rho$ by $\lambda|_Z \nu_\rho$. This follows from the scalar action and does not require division by the degree of ρ .

Suppose a group A acts on the primitive central idempotents of kG , and there is an injective homomorphism $a \mapsto \lambda_a$ from A to the linear characters of G . Assume every λ_a is trivial on N . Suppose every block contains a finite dimensional simple representation, and tensoring any such representation in b by λ_a gives a representation in $a \cdot b$.

Theorem 7.20 (Stabilisers under tensoring by linear characters). *Under these assumptions, the stabiliser $A_b = \{a \in A : a \cdot b = b\}$ is finite. For every $a \in A_b$, the character λ_a factors through $G/NZ(G)$, and*

$$|A_b| \leq [G : NZ(G)].$$

Proof. The central ℓ -regular elements form a subgroup $Z_{\ell'} \leq Z(G)$ of order prime to ℓ . Choose a simple representation ρ belonging to b . If $a \in A_b$, then $\lambda_a \otimes \rho$ belongs to the same block. Proposition 7.6 gives equality of their scalar characters on $Z_{\ell'}$. Since twisting multiplies that character by $\lambda_a|_{Z_{\ell'}}$, cancellation in $k^\times$ proves that λ_a is trivial on $Z_{\ell'}$.

It is also trivial on every ℓ -element. If $g^{\ell^m} = 1$, then $\lambda_a(g)^{\ell^m} = 1$. In characteristic ℓ , $X^{\ell^m} - 1 = (X - 1)^{\ell^m}$, so a field has no nontrivial ℓ -power roots of unity. Every central element is the product of its commuting ℓ -part and ℓ -regular part, both powers of the original element. Hence λ_a is trivial on all of $Z(G)$, and therefore on $NZ(G)$.

Thus λ_a induces a character $\bar{\lambda}_a : G/NZ(G) \rightarrow k^\times$ with $\lambda_a = \bar{\lambda}_a \circ \pi$. These quotient characters are distinct, since composing with π recovers λ_a and $a \mapsto \lambda_a$ is injective. These quotient characters form a linearly independent family in the space of all functions on $G/NZ(G)$, which has dimension $[G : NZ(G)]$. Therefore their indexing set A_b is finite and $|A_b| \leq [G : NZ(G)]$. $\square$

Implementation.

The assumptions on simple representations and tensoring are expressed by `ModularRep.LinearBlockStabilizer.SimpleBlockTensorSource`. It supplies a simple representation in every block and compatibility between the block action and tensoring. The theorem is `ModularRep.LinearBlockStabilizer.modular_block_stabilizer_bound`. The quotient character is defined by `descendCenterQuotient`. Mathlib's linear independence of multiplicative characters gives `finite_of_injective_linear_characters` and the bound `card_le_of_injective_linear_characters`, in the same namespace. The theorem `block_stabilizer_finite` explicitly supplies finiteness of the stabiliser for both the ordinary and modular character actions.

Example 7.21 (All quotient characters). Take A to be the linear characters of G trivial on N , identified with those of G/N . Once their block action and its compatibility with tensoring are given, every character stabilising a block factors through $G/NZ(G)$, and the stated bound follows. In the special case $NZ(G) = G$, the bound is one, so every block has trivial stabiliser for this action.

For the ordinary form, let K be a field of characteristic zero. Set $m = |G|_{\ell'}$ and fix an isomorphism $\mu_m(k) \simeq \mu_m(K)$ between the groups of m th roots of unity. Applied pointwise, it

identifies ordinary linear characters of G/N of order prime to ℓ with modular linear characters of G/N . Transport the block action through this identification. The stabilisers are isomorphic, and preservation of kernels gives the factorisation through $G/NZ(G)$ and the same bound.

Implementation.

`ModularRep.LinearBlockStabilizerReduction` constructs `ordinaryToBrauerQuotient`, `reductionStabilizerEquiv`, and `ordinary_block_stabilizer_bound` in namespace `ModularRep.LinearBlockStabilizer`. The theorem `reduction_commutes_with_tensor` requires a modular system, `StableReductionBrauerCharacterCompatibility` and `BrauerLinearTensorProductFormula`.

Chapter 8

Weights and changes of representatives

Let G be a finite group and let ℓ be a prime. A weight is a local representation associated with an ℓ -subgroup of G . The modular formulation uses simple modular representations that are projective over the normaliser quotient [1, p. 369]. We use the ordinary character formulation. See, for example, [6, p. 90].

Throughout this chapter, K is a field of characteristic zero. Write $\text{Irr}_K(H)$ for the trace characters of irreducible finite dimensional K -representations of a finite group H . The weight definitions below use these trace characters. They agree with the usual ordinary character definition of weights when K is a splitting field for the local quotients $N_G(Q)/Q$.

8.1 Weights and their local representations

Definition 8.1. A character $\theta \in \text{Irr}_K(H)$ has *defect zero* if it is afforded by a simple KH -module V such that

$$(\dim_K V)_\ell = |H|_\ell.$$

Denote the set of these characters by $\text{dz}_\ell(H)$. Since $\theta(1) = \dim_K V$, the numerical condition is equivalent to $\theta(1)_\ell = |H|_\ell$.

In *character form*, a weight of G is a pair (Q, θ) , where Q is an ℓ -radical subgroup of G and

$$\theta \in \text{dz}_\ell(N_G(Q)/Q).$$

In *representation form*, a weight is a pair (Q, V) with the same condition on Q , where V is a simple finite dimensional $K[N_G(Q)/Q]$ -module of defect zero.

At a fixed subgroup Q , two weights in representation form are identified when their local representations are isomorphic. Two weights in character form are identified when their local character functions are equal. The condition that Q is radical is included in these definitions.

Proposition 8.2 (Comparison by characters). *Taking characters gives a bijection from weights in representation form up to local isomorphism to weights in character form. It induces an $\text{Aut}(G)$ -equivariant bijection on G -conjugacy classes of weights.*

Proof. Trace is invariant under isomorphism, so the map is well defined. Character separation for finite groups over fields of characteristic zero gives injectivity, as in Section 3.1. A simple representation affording a character of defect zero gives surjectivity. Transport along a group isomorphism preserves dimension and simplicity, and taking traces commutes with this transport. The bijection therefore commutes with automorphisms and descends to conjugacy classes. $\square$

Implementation.

The local representations use Mathlib's `FDRep`, with `FDRep.char_iso` giving invariance of the trace under isomorphism. The Lean comparison maps take `ModularRep.LocalSplittingCharacterInput` in `ModularRep.WeightCharacterBridge` as an explicit argument. Its fields require that the trace of every simple local representation has an irreducible realisation and that equal traces determine simple local representations up to isomorphism. These properties follow mathematically from the definition of Irr_K and the character separation argument in Section 3.1. The record does not require K to be a splitting field. The character map is `ModularRep.LocalSplittingCharacterInput.characterOfSimple`. The two bijections are `ModularRep.CharacterWeight.isoClassEquiv` and `ModularRep.CharacterWeight.conjugacyClassEquiv`, for `ModularRep.CharacterWeight`.

8.2 Automorphisms and conjugacy classes

An isomorphism $\phi : H \rightarrow H'$ transports a representation by composition with ϕ^{-1} . Its character is therefore $\theta \circ \phi^{-1}$. This is also the convention for transport of class functions in [6, Problem 2.1]. In particular, $\alpha \in \text{Aut}(G)$ sends a weight at Q to a weight at $\alpha(Q)$ through the induced isomorphism

$$\bar{\alpha} : N_G(Q)/Q \longrightarrow N_G(\alpha(Q))/\alpha(Q).$$

The transported local character is $\theta \circ \bar{\alpha}^{-1}$. Composition of these transports gives the usual left action of $\text{Aut}(G)$. The corresponding right action uses α^{-1} , so its subgroup coordinate is $\alpha^{-1}(Q)$.

Implementation.

`ModularRep.OrdinaryIrreducibleCharacter.mapEquiv_apply` gives the formula $\theta(\phi^{-1}h')$. Representation transport uses Mathlib's restriction functor `Action.res` and the equivalence `Action.resEquiv`. `ModularRep` constructs the induced normaliser quotient maps and proves that the resulting weight transports preserve radicality and defect zero. The actions on weights in representation and character form are defined before passing to their classes.

Conjugation in G gives an equivalence relation on weights. Write $\text{Alp}(G)$ for the set of weights in character form modulo this relation. Automorphisms act on $\text{Alp}(G)$ because they preserve conjugacy. Local isomorphism and conjugacy in G have different roles: local isomorphism keeps Q fixed, whereas conjugation may replace it by another subgroup.

Proposition 8.3 (Weights with a fixed radical subgroup). *Fix an ℓ -radical subgroup $Q \leq G$. Let $\text{Alp}(G)_{[Q]}$ be the set of G -conjugacy classes of weights in character form whose radical subgroup is conjugate to Q . Then*

$$\text{dz}_\ell(N_G(Q)/Q) \longrightarrow \text{Alp}(G)_{[Q]}, \quad \theta \longmapsto [(Q, \theta)]$$

is a bijection.

Proof. Every class in $\text{Alp}(G)_{[Q]}$ has a representative at Q , obtained by conjugating its radical subgroup to Q . This proves surjectivity.

If (Q, θ) and (Q, η) are conjugate by $g \in G$, then $g \in N_G(Q)$. The induced automorphism of $N_G(Q)/Q$ is conjugation by gQ . Characters are constant on conjugacy classes, so it fixes θ . Hence $\eta = \theta$, proving injectivity. The same argument shows that the character obtained by transporting to Q is independent of the chosen conjugating element. $\square$

Implementation.

The bijection is `ModularRep.CharacterWeight.localDefectZeroEquivWeightRadicalFibre`. Its restriction to a block is `representativeDZEquivWeightBlockRadicalFibre` in the same namespace, with the local block induction hypotheses described below.

8.3 The block of a weight

Fix compatible ordinary and modular character conventions for the local groups, including the splitting and root identifications needed for their block assignments. Let (Q, θ) be a weight. Inflate θ to $N_G(Q)$ and let b be its block. The weight belongs to an ambient block B when Brauer block induction is defined and $b^G = B$ [6, p. 90].

Recall the definition of this induction. If $H \leq G$ and $\lambda_b : Z(kH) \rightarrow k$ is the central character of a block b , extend it linearly to $Z(kG)$ by

$$\lambda_b^G(C^+) = \lambda_b((C \cap H)^+)$$

for each conjugacy class C of G . Here C^+ denotes the sum of the elements of C in the group algebra, and k is algebraically closed of characteristic ℓ . If λ_b^G is an algebra homomorphism, it is the central character of a unique block, denoted by b^G . This is Brauer's definition of induced block [6, p. 87]. For an ℓ -subgroup P , the inclusions

$$PC_G(P) \leq H \leq N_G(P)$$

ensure that induction is defined for every block of H [6, Theorem 4.14].

Write $\text{Bl}_\ell(G)$ for the set of ℓ -blocks of G . Suppose the local block assignment, inflation and block induction commute with automorphisms. Suppose also that inner automorphisms fix ambient blocks. Then the ambient block assigned to a weight depends only on its G -conjugacy class, and the resulting map $\text{Alp}(G) \rightarrow \text{Bl}_\ell(G)$ is equivariant. The corresponding compatibility of Brauer block induction with group isomorphisms is stated in [6, Problem 4.4].

Implementation.

`ModularRep.CharacterWeight.LocalBlockInductionOperations` in `ModularRep.CharacterWeightBlockAssignment` supplies the local character block and inflation. It also supplies finite complete block decompositions of G and the relevant normalisers, together with their block central characters, and multiplicativity of the induced central function. These data determine the ambient block. `ModularRep.CharacterWeight.LocalBlockInductionSource` adds compatibility with automorphisms and invariance under inner automorphisms. Its maps `LocalBlockInductionSource.weightBlock` and `LocalBlockInductionSource.weightBlock_transport` give the assignment on conjugacy classes and its equivariance. The identification of the supplied operations with ordinary and modular block theory is a hypothesis.

Example 8.4 (Local maps from a global correspondence). Let $\Omega : X \rightarrow \text{Alp}(G)$ be a bijection. For each representative Q of a radical conjugacy class, set

$$X_Q = \{x \in X : \Omega(x) \in \text{Alp}(G)_{[Q]}\}.$$

These subsets partition X . Restricting Ω and applying Proposition 8.3 gives a bijection

$$X_Q \longrightarrow \text{dz}_\ell(N_G(Q)/Q).$$

If Ω is equivariant, these local maps commute with transport of subgroups and characters. The construction for Brauer characters with a fixed compatible root embedding is in `ModularRep.PaperProofs.IntrinsicGlobalToRepresentativeMaps`.

8.4 Quotient by the ℓ -core

Proposition 8.5 (Transport through the normal ℓ -core). *Let $P = O_\ell(G)$. A weight (Q, V) of G in representation form determines a weight in representation form of G/P with radical subgroup Q/P and local representation transported along*

$$N_G(Q)/Q \simeq N_{G/P}(Q/P)/(Q/P).$$

Conversely, a weight in representation form of G/P determines one of G by taking the full inverse image of its radical subgroup and transporting the local representation along the inverse isomorphism. Both constructions preserve the dimension of the local representation.

Proof. By Proposition 2.2, every radical subgroup Q contains P . Proposition 2.4 identifies radicality before and after quotienting by P . The normaliser quotient isomorphism then transports the local representation. An isomorphism of groups preserves simplicity, dimension and group order, and hence preserves defect zero. The same reasoning applies to the full inverse image of a radical subgroup of G/P . $\square$

Implementation.

The two constructions are `ModularRep.RepresentationWeight.quotientPCore` and `ModularRep.RepresentationWeight.liftFromQuotientPCore` in `ModularRep.NormalCoreTransport`. They construct individual weights. An inverse equivalence on conjugacy classes is a separate assertion.

A comparison for weights in a specified block also requires compatibility of the character conventions and block assignments on the two local quotients. For applications involving modular character triples, the required triple relation must hold for the transported local character, with the specified roots and block operations.

Chapter 9

Basic sets and equivariant bijections

An integral basic set identifies two free abelian groups, one indexed by ordinary characters and the other by Brauer characters. If a group acts compatibly with decomposition, this identification is equivariant. For a hypoelementary group, Conlon's theorem and injectivity of the mark homomorphism give a bijection of the two sets of characters. Further results in this chapter extend equivariant maps from stabilisers, cancel invariant subsets and construct fixed representatives in product orbits.

Write A for an acting group. An A -equivariant map satisfies $f(ax) = af(x)$, and A_x denotes the stabiliser of x . If a map $\pi : X \rightarrow I$ is given, its fibre over i is the subset $X_i = \{x : \pi(x) = i\}$.

9.1 Naturality and an integral basic set

Let G be finite and let ℓ be prime. Let K and k be fields of characteristic zero and ℓ , respectively. Consider the Grothendieck groups $K_0(KG)$ and $K_0(kG)$ of Chapter 5, and a specified decomposition homomorphism $d : K_0(KG) \rightarrow K_0(kG)$. Let X and Y be finite sets. For a set X , let $\mathbb{Z}X$ denote the free abelian group with basis $(e_x)_{x \in X}$. Suppose X and Y index distinct simple modules over KG and kG , respectively. Sending a basis vector to the corresponding module class gives injective maps $j_X : \mathbb{Z}X \rightarrow K_0(KG)$ and $j_Y : \mathbb{Z}Y \rightarrow K_0(kG)$, by the basis of simple classes in Theorem 5.7.

Definition 9.1. The family indexed by X is an *integral basic set* for the family indexed by Y if the restriction of d is an isomorphism $j_X(\mathbb{Z}X) \rightarrow j_Y(\mathbb{Z}Y)$. Equivalently, there is an isomorphism $D : \mathbb{Z}X \rightarrow \mathbb{Z}Y$ such that $j_Y D = d j_X$. If A acts on X and Y , it acts on the free groups by $ae_x = e_{ax}$ and $ae_y = e_{ay}$. Compatible actions on the two Grothendieck groups are actions for which j_X and j_Y commute with A .

Proposition 9.2 (Equivariance of the restricted decomposition map). *Suppose these actions are compatible and d commutes with A . Then D is an isomorphism of integral permutation lattices. If $D(e_x) = \sum_y d_{y,x} e_y$, its coefficients satisfy $d_{ay,ax} = d_{y,x}$ for every $a \in A$.*

Proof. For $v \in \mathbb{Z}X$, use the defining square and naturality to obtain

$$j_Y D(av) = dj_X(av) = d(aj_X v) = adj_X v = aj_Y Dv = j_Y(aDv).$$

Injectivity of j_Y gives $D(av) = aD(v)$. Taking coefficients on basis vectors proves the stated matrix identity. Conversely, that identity on basis vectors implies equivariance by linearity. $\square$

The next section gives conditions under which the two indexing sets are also equivariantly bijective. For automorphism twisting, the construction in Chapter 6 gives naturality under its stated compatibility hypothesis. For tensoring with linear characters, the ordinary and modular actions must satisfy the corresponding character and root compatibility laws.

Implementation.

The permutation representations are Mathlib's `Representation.ofMulAction`, with coefficients in $\mathbb{Z}$. `ModularRep.DecompositionBasicSetBridge` defines `RestrictedIntegralBasicSet` and proves `matrixEquivariant_of_kZero_naturality`. Its `matrixEquivariant_of_stableReduction` obtains naturality from the constructed decomposition map. The equivalence between the matrix and linear map formulations is in `ModularRep.IntegralBasicSetBridge`.

Proposition 9.3 (Restriction to an invariant block). *Suppose D is equivariant and $\pi_X : X \rightarrow I$, $\pi_Y : Y \rightarrow I$ are A -invariant block assignments. If*

$$D(\mathbb{Z}X_i) \subseteq \mathbb{Z}Y_i, \quad D^{-1}(\mathbb{Z}Y_i) \subseteq \mathbb{Z}X_i \quad (i \in I),$$

then D restricts to an isomorphism of $\mathbb{Z}A$ -lattices $\mathbb{Z}X_i \rightarrow \mathbb{Z}Y_i$ for every $i \in I$.

Proof. The support conditions restrict D and its inverse to the two submodules spanned by the indicated fibres. The restrictions are inverse because the original maps are inverse. The action preserves each submodule, and the restricted maps still commute with it. Identifying each supported submodule with the free group on its fibre gives the assertion. $\square$

Implementation.

`ModularRep.BlockFibreRestriction`, especially `restrictBlockLinearEquiv`, `restrictBlockLinearEquiv_equivariant` and `restrictRestrictedIntegralBasicSet`. Both support conditions are hypotheses.

9.2 From permutation lattices to finite sets

Let p be a prime, and write $\mathbb{Z}_p$ for the ring of p -adic integers. The prime p may differ from the characteristic ℓ used for modular characters. A finite group A is *p -hypoelementary* if it has a normal p -subgroup with cyclic quotient of order prime to p . For an A -set X and a subgroup $U \leq A$, the *mark* at U is $|X^U|$, where

$$X^U = \{x \in X : ux = x \text{ for every } u \in U\}.$$

Theorem 9.4 (Conlon). *Let A be a finite group and let X, Y be finite A -sets. If $\mathbb{Z}_p X$ and $\mathbb{Z}_p Y$ are isomorphic as $\mathbb{Z}_p A$ -lattices, then*

$$|X^U| = |Y^U|$$

for every p -hypoelementary subgroup U of A .

Conlon's fixed coset calculation gives this implication [4, Section 4, especially (4.7)]. Bouc gives a modern formulation, including the converse, in [3, Theorem 3.5.5]. In that account the coefficient ring is a complete commutative Noetherian local ring with residue characteristic p , and lattices are finitely generated and free over that ring. The choice $\mathbb{Z}_p$ satisfies these hypotheses.

Theorem 9.5 (Burnside's determination by marks). *Let A be a finite group. Two finite A -sets X and Y are isomorphic if and only if $|X^U| = |Y^U|$ for every subgroup $U \leq A$.*

Proof. An equivariant bijection identifies the fixed points of every subgroup. For the converse, decompose each set into transitive orbits A/H , with H taken from a set of representatives of subgroup conjugacy classes. The matrix

$$m(U, H) = |(A/H)^U|$$

is triangular when those representatives are ordered by subgroup order. Its diagonal entries are $|N_A(H) : H|$, which are nonzero. Equality of marks therefore gives equality of the multiplicities of all transitive orbits. $\square$

For this statement and proof, see, for example, [3, Theorem 2.3.2]. The equivalent formulation is injectivity of the mark homomorphism from the Burnside ring.

Corollary 9.6 (Permutation lattices). *Let A be a finite p -hypoelementary group and let X, Y be finite A -sets. If $\mathbb{Z}_p X \simeq \mathbb{Z}_p Y$ as $\mathbb{Z}_p A$ -lattices, then X and Y are equivariantly bijective. In particular, the same conclusion holds if $\mathbb{Z} X \simeq \mathbb{Z} Y$ as $\mathbb{Z} A$ -lattices.*

Proof. Conlon's theorem gives equality of the marks at all p -hypoelementary subgroups. Every subgroup of A is p -hypoelementary by Proposition 2.7. Thus all marks agree, and the preceding theorem gives the bijection. For the integral version, extend the isomorphism and its inverse from $\mathbb{Z}$ to $\mathbb{Z}_p$. $\square$

Implementation.

`ModularRep.PaperProofs.ConlonBasicSet` constructs scalar extension and the mark map on isomorphism classes of finite A -sets. The scalar extension uses Mathlib's equivalences for tensor products with finitely supported functions. The intertwining identities are proved in this module. The two published theorems are separate hypotheses: `PadicConlonMarkDetection` and `PublishedBurnsideMarkInjectivity`. Both range over all finite A -sets. Under these hypotheses, `corollary_2_4_conlonMark` proves the implication from isomorphic p -adic permutation lattices to an equivariant bijection. The proofs of the published theorems are not part of that declaration.

If A is p -hypoelementary, Proposition 9.2 and Corollary 9.6 give an equivariant bijection between the ordinary basic set and the modular characters it indexes. Under the same hypothesis, Proposition 9.3 gives the corresponding bijection on an invariant block.

Implementation.

The corresponding declarations are `equivariantSetEquiv_of_kZero_naturality` and `equivariantSetEquiv_restrictBlock_of_kZero_naturality` in the modules for basic sets and block restriction. They assume the specified integral basic set, actions, naturality, hypoelementary group and published mark theorems. They prove existence of a bijection without prescribing its pairs. No unitriangularity hypothesis is used.

9.3 Extending a bijection over an orbit

Let $\pi_X : X \rightarrow I$ and $\pi_Y : Y \rightarrow I$ be equivariant maps, and assume I is transitive.

Proposition 9.7 (Extension from one fibre). *Fix $i_0 \in I$ and an A_{i_0} -equivariant bijection $\beta_0 : X_{i_0} \rightarrow Y_{i_0}$. There is a unique A -equivariant bijection $\beta : X \rightarrow Y$ that preserves the maps to I and restricts to β_0 on X_{i_0} .*

Proof. For each $i \in I$, choose $t_i \in A$ with $t_i i_0 = i$, and define

$$\beta(x) = t_i \beta_0(t_i^{-1} x) \quad (x \in X_i).$$

The expression has image in Y_i . Replacing β_0 by its inverse gives an inverse map on every fibre and hence on the total sets. At $i = i_0$, the chosen t_{i_0} belongs to A_{i_0} , so equivariance of β_0 shows that the restriction is exactly β_0 .

Two choices of t_i differ by an element of A_{i_0} . Equivariance of β_0 under that stabiliser makes the formulas identical. For $a \in A$, the element $t_{ai}^{-1}at_i$ also fixes i_0 . Applying A_{i_0} -equivariance to this element proves $\beta(ax) = a\beta(x)$. Finally, an equivariant extension of β_0 must send $t_i(t_i^{-1}x)$ to $t_i\beta_0(t_i^{-1}x)$, which proves uniqueness. $\square$

Implementation.

`Formalisation.FibreTransport`, with `Formalisation.transportedEquiv`, `Formalisation.transportedEquiv_equivariant`, `Formalisation.transportedEquiv_independent_of_transporter` and `Formalisation.transportedEquiv_preserves_base`. The equivalence of the total sets uses Mathlib's `Equiv.ofFiberEquiv`. The transporter formula, its equivariance and its independence of the choices are proved in this module. The uniqueness assertion follows from the formula for β , as in the proof above. It is not a separate theorem in this module.

For an arbitrary A -set I , choose one representative i of each orbit and an A_i -equivariant bijection $X_i \rightarrow Y_i$. Applying the proposition on each orbit gives an A -equivariant bijection $X \rightarrow Y$ preserving the maps to I . The module `Formalisation.IndexedAssembly` combines families satisfying $\beta_{ai}(ax) = a\beta_i(x)$. The transporter formula above proves precisely this compatibility from the bijections on the representative fibres.

The same formula preserves any specified A -invariant relation $\mathcal{R} \subseteq X \times Y$, provided the initial bijections satisfy it. More explicitly, assume

$$\begin{aligned} (x, y) \in \mathcal{R} &\implies (ax, ay) \in \mathcal{R} & (a \in A), \\ (x, \beta_i(x)) &\in \mathcal{R} & (x \in X_i) \end{aligned}$$

with the second condition imposed on each representative fibre. Every $x \in X$ has the form au for some u in one of those fibres, and $(x, \beta(x)) = (au, a\beta_i(u)) \in \mathcal{R}$. Thus the local relation and its invariance are separate hypotheses in an application.

Example 9.8 (A block correspondence on an orbit). Suppose I is an orbit of blocks, X_i is its Brauer character set and Y_i its set of weight classes. An equivariant bijection for one block, under that block's stabiliser, extends to the whole orbit by Proposition 9.7. It still sends each X_i into Y_i . A modular character triple relation is preserved by the extension provided it holds for each pair on the chosen fibre and is invariant under the automorphisms being used.

9.4 Involutions and cancellation

Proposition 9.9 (Classification of finite sets with an involution). *Let σ and τ be involutions on finite sets X and Y . There is a bijection $\beta : X \rightarrow Y$ with $\beta\sigma = \tau\beta$ if and only if $|X| = |Y|$ and $|X^\sigma| = |Y^\tau|$.*

Proof. Each orbit of an involution has size one or two. If f is the number of fixed points and t the number of orbits of size two, then $|X| = f + 2t$. Thus the two stated equalities imply that the numbers of orbits of each size agree. Match the singleton orbits and the orbits of size two. A choice of one point in each paired orbit of size two determines the bijection on its other point by equivariance. Combining these maps gives β . Conversely, any such β preserves total size and carries the fixed points bijectively to the fixed points. $\square$

Equivalently, the two counts determine the cycle type of an involution and hence its conjugacy class in the symmetric group.

Implementation.

Formalisation. `C2Cancellation`, including `card_eq_fixed_add_twice_cycleCount` and `exists_equivariantEquiv_of_card_eq_of_fixed_card_eq` in

Formalisation. `C2Cancellation`. These results apply Mathlib's fixed point counts and classification of permutations by cycle type, `Equiv.Perm.isConj_iff_cycleType_eq`, to involutions.

The pair $(|X|, |X^\sigma|)$ determines a finite set with involution up to equivariant bijection. Both entries add under invariant disjoint unions. Consequently, if $X \simeq Y$ equivariantly and invariant subsets $X' \subseteq X, Y' \subseteq Y$ are equivariantly bijective, then $X \setminus X' \simeq Y \setminus Y'$ equivariantly. The argument constructs a bijection of the complements from their two counts.

For two parts exchanged by the involution, a different formula is useful. Choose a bijection $f : X_0 \rightarrow Y_0$, and define the map on X_1 by $\tau f \sigma$. It takes X_1 to Y_1 , is inverse to the analogous map constructed from f^{-1} and commutes with the involutions. These constructions are in `Formalisation.BlockCancellation` and `Formalisation.C2Cancellation`.

Proposition 9.10 (Cancellation over a block orbit). *Let I be a finite set with an involution, and let $X \rightarrow I$ and $Y \rightarrow I$ be equivariant maps of finite sets with involutions. Write $I_0, \dots, I_r$ for the orbits in I , and let U_j and V_j be their inverse images in X and Y . Suppose that $X \simeq Y$ equivariantly and that $U_j \simeq V_j$ equivariantly for $1 \leq j \leq r$. Then there is an equivariant bijection $U_0 \rightarrow V_0$ preserving the map to I_0 .*

Proof. Cancel the known parts in the total size and fixed point count. Proposition 9.9 gives an equivariant bijection between the remaining parts. If I_0 is a singleton, it already preserves the map to I_0 .

Otherwise write $I_0 = \{i, \sigma i\}$. The involutions exchange the two fibres on each side, so each fibre has half the size of its corresponding remaining part. These sizes agree. Choose a bijection $f : X_i \rightarrow Y_i$ and define the map on the other fibre by $\tau f \sigma$, where σ and τ denote the source and target involutions. The two formulas give an equivariant bijection preserving both fibres. $\square$

For an application to blocks, the maps to I assign a block to each character or weight class. The hypotheses then require equivariant comparisons on each of the other block orbits.

Implementation.

In `Formalisation.BlockCancellation`, `cancel_equivariant_equiv_of_parts` handles the finite disjoint union of known parts, and `cancel_exchanged_blocks_preserving` constructs the map on the remaining pair of exchanged fibres. The resulting map need not be the restriction of the initially given equivalence $X \simeq Y$.

Example 9.11. A set with eight elements and two fixed points has three orbits of size two. If an invariant part consists of one such orbit, the complement has the pair of counts $(6, 2)$. Any other complement with these counts is isomorphic as a set with involution.

9.5 Fixed representatives in a product

Let $D \rtimes R$ act on a set X . Suppose that

$$(D \rtimes R)_x = D_x \rtimes R_x$$

for some $x \in X$. If $r \in R$ satisfies $rx \in D_x$, then $rx = x$. Indeed, write $rx = dx$. The element (d^{-1}, r) fixes x , so the stabiliser equality implies $r \in R_x$.

Proposition 9.12 (A fixed tuple in a product orbit). *Let $n > 0$, let $O_0, \dots, O_{n-1}$ be D -orbits in X , and let $T : X \rightarrow X$. Suppose a cyclic group $E = \langle \tau \rangle$ acts on X^n by*

$$\tau(y_0, \dots, y_{n-1}) = (Ty_{n-1}, Ty_0, \dots, Ty_{n-2}).$$

Assume that τ preserves $\prod_j O_j$. Choose $x \in O_0$ such that

$$T^n x = rx \quad \text{for some } r \in R, \quad (D \rtimes R)_x = D_x \rtimes R_x.$$

Then $(x, Tx, \dots, T^{n-1}x)$ belongs to $\prod_j O_j$ and is fixed by E .

Proof. Varying the j th coordinate of a tuple in the product shows that $T(O_j) \subseteq O_{j+1}$, with indices taken modulo n . Consequently $T^n x \in O_0 = D_x$. The preceding observation gives $rx = x$, and hence $T^n x = x$. Successive transport places $T^j x$ in O_j . The formula defining τ now shows that it fixes the tuple, so its cyclic group E fixes the tuple as well. $\square$

Apply this construction on each cycle when there are several cycles of coordinates. The groups D and R may vary from cycle to cycle. If the actions extend compatibly to $D_{\text{prod}} \rtimes E$, where D_{prod} is the product of the coordinate groups, the resulting tuple θ satisfies

$$(D_{\text{prod}} \rtimes E)_\theta = (D_{\text{prod}})_\theta \rtimes E.$$

Indeed, every element of E fixes θ , and (d, e) fixes θ exactly when d does.

Implementation.

`ModularRep.ComponentReturnFull` allows separate groups and orbit data on every cycle. The theorem `component_return_full_of_product_orbit_stable` in `ModularRep.ManuscriptVerification.ComponentReturnFull` derives the conditions on coordinate orbits from stability of the product orbit and constructs the fixed tuple. In a representation theoretic application, the product character sets, their diagonal actions and the action after a full cycle must satisfy the stated action formulas.

Chapter 10

Character counts from finite matrices

Restriction of ordinary characters to regular elements gives a matrix whose rank is the number of irreducible Brauer characters. If an involution fixes the block, a second rank calculation gives the number of Brauer characters it fixes. Both calculations follow from linear algebra once the matrix entries and the action on conjugacy classes are identified.

10.1 The rank of ordinary restrictions

Lemma 10.1. *Let M, N, V be vector spaces over a field K , with N finite dimensional. Let $D : M \rightarrow N$ and $E : N \rightarrow V$ be linear maps. If D is surjective and E is injective, then*

$$\dim_K \operatorname{im}(ED) = \dim_K N.$$

Proof. Surjectivity gives $\operatorname{im}(ED) = \operatorname{im} E$. Injectivity identifies $\operatorname{im} E$ with N . □

We recall the character theoretic application over the complex numbers, with a fixed choice of roots for Brauer characters. For an ℓ -block b , write $\operatorname{Irr}(b)$ and $\operatorname{IBr}(b)$ for its ordinary irreducible characters and irreducible Brauer characters.

Theorem 10.2 (Rank of ordinary restrictions). *Let b be an ℓ -block of a finite group G . If χ^0 denotes restriction to the ℓ -regular elements, then*

$$\dim_{\mathbb{C}} \langle \chi^0 : \chi \in \operatorname{Irr}(b) \rangle_{\mathbb{C}} = |\operatorname{IBr}(b)|. \quad (10.1)$$

Proof. The irreducible Brauer characters are linearly independent. The decomposition formula is

$$\chi^0 = \sum_{\varphi \in \operatorname{IBr}(b)} d_{\chi\varphi} \varphi \quad (\chi \in \operatorname{Irr}(b)).$$

The decomposition matrix of b has rank $|\operatorname{IBr}(b)|$. Thus its rows span the coordinate space on $\operatorname{IBr}(b)$. Apply the lemma with D sending the ordinary character basis to these rows and E sending the coordinate basis to the Brauer characters. □

For linear independence, the decomposition formula and full rank of the decomposition matrix, see, for example, [6, Theorem 2.6 and Corollaries 2.9–2.11]. The passage to one block follows from the block diagonal form of the decomposition matrix [6, Theorem 3.3 and the following discussion]. The related equality between $|\operatorname{IBr}(G)|$ and the number of regular conjugacy classes is Corollary 2.10 in that account.

The same rank argument works over a field K of characteristic zero if the characters take values in K , the decomposition formula holds there, the Brauer characters are linearly independent over K , and the decomposition rows span the coordinate space over K . Theorem 4.11 gives independence under its stated field and root hypotheses. The decomposition and spanning conditions are additional hypotheses.

Implementation.

Formalisation. `RestrictionRank` proves `finrank_range_comp_eq_card` and `finrank_span_matrix_combinations_eq_card`. The range calculation applies Mathlib's `LinearMap.range_comp_of_range_eq_top` and `LinearMap.finrank_range_of_inj`, followed by its dimension formula for a finite coordinate space. The second theorem assumes independent Brauer vectors and decomposition rows spanning their coordinate space. It proves the linear algebra implication over K . The identification of these vectors and coefficients with characters and decomposition numbers is separate.

Let $c_1, \dots, c_t$ represent all ℓ -regular conjugacy classes. Evaluation at these representatives is injective on class functions defined on the ℓ -regular elements. Consequently the matrix

$$M = (\chi(c_j))_{\chi \in \text{Irr}(b), 1 \leq j \leq t}$$

has row rank $|\text{IBr}(b)|$. More generally, one may use any list meeting every regular conjugacy class. To apply the formula to a specified matrix, its rows must be the indicated ordinary characters and its columns must evaluate those characters at such a list.

10.2 Fixed points from symmetrised rows

Proposition 10.3 (An involution permuting a basis). *Let K have characteristic zero. Let $(v_i)_{i \in I}$ be a finite basis of a K -vector space V , and let $T : V \rightarrow V$ be a linear map such that $T(v_i) = v_{\sigma(i)}$ for an involution σ of I . Set*

$$r = \dim_K V, \quad s = \dim_K \text{im}(1 + T).$$

Then $|I^\sigma| = 2s - r$.

Proof. The permutation matrix of T has diagonal entry one at a fixed index and zero otherwise. Hence $\text{tr}(T) = |I^\sigma|$ in K . Since $T^2 = 1$, the map $P = (1 + T)/2$ is a projection onto $\text{im}(1 + T)$. The trace of a projection is its rank, so

$$s = \text{tr}(P) = \frac{1}{2}(r + \text{tr}(T)).$$

Characteristic zero gives the asserted equality of integers. □

The basis need not be known explicitly. Suppose $(g_a)_{a \in A}$ spans V and $E : V \rightarrow K^C$ is an injective linear map. Let M be a matrix and let π permute C , with

$$E(g_a)(c) = M_{a,c}, \quad E(Tg_a)(c) = M_{a,\pi(c)}.$$

The vectors $(1 + T)g_a$ span $\text{im}(1 + T)$. Their images under E are the rows of the *symmetrised matrix*

$$M_{a,c}^+ = M_{a,c} + M_{a,\pi(c)}.$$

Thus M has row rank r and M^+ has row rank s .

For a block fixed by an involution in $\text{Aut}(G)$, take the basis to be its irreducible Brauer characters and the spanning vectors to be its ordinary restrictions. Once π describes the action on the chosen regular class representatives, the two ranks give

$$(|\text{IBr}(b)|, |\text{IBr}(b)^\sigma|) = (r, 2s - r).$$

The action formula must use the same characters, roots and class representatives as the evaluation formula.

Implementation.

The trace and projection argument is `fixed_card_add_card_eq_two_mul_plus_rank` in `ModularRep.PaperProofs.SporadicFi24P3Definition44NamedCarrierInvolutionBasisRank`. It applies to any finite basis over a field of characteristic zero. Its proof uses Mathlib's `Matrix.trace_permutation` and `LinearMap.IsProj.trace`. The local argument identifies the basis action with a permutation matrix and relates the projection to the image of $1 + T$. The comparison with the symmetrised row rank is `finrank_range_id_add_eq_symmetrizedRows` in `ModularRep.PaperProofs.SporadicFi24P3Definition44NamedCarrierSymmetrizedRowRankEquality`. Its hypotheses are the spanning, injectivity and evaluation identities above. The matrix identity in this latter theorem allows any map on the column index set.

10.3 Certifying a matrix rank

Proposition 10.4 (A minor and spanning rows). *Let M be a finite matrix over a field K . Suppose a square minor on r selected rows and r selected columns has nonzero determinant. If every row of M is a linear combination of the selected rows, then M has row rank r .*

Proof. Restricting the selected rows to the selected columns gives the rows of an invertible matrix, so they are linearly independent. By hypothesis they span the row space of M . They therefore form a basis of that space. $\square$

Implementation.

`ModularRep.PaperProofs.FiniteRowRankCertificate` records the selected rows and columns, the determinant of the minor, its nonvanishing and the coefficients expressing every row as a combination of the selected rows. `FiniteRowRankCertificate.rows_finrank` proves the rank from the determinant identity and the equations expressing every row in terms of the selected rows. It uses Mathlib's `Matrix.linearIndependent_rows_of_det_ne_zero` for the selected minor and `finrank_span_eq_card` for the dimension of the span. The local proof shows that the selected rows span the full row space.

Together with the evaluation and action identities, these finite equations give character counts and counts of fixed characters. To use the rank calculation for character counts, one must identify the full matrix with the restrictions of all the intended characters to all the relevant regular classes. The selected minor proves that the selected rows are linearly independent, and the given row expressions prove that they span the full row space.

10.4 Finite scalar products and character selection

Let K be a field of characteristic zero, let H be a finite group, and let $(C_j)_{j \in J}$ be distinct conjugacy classes with representatives u_j , where J is finite. If a function $f : H \rightarrow K$ is constant on each C_j and vanishes outside their union, partitioning the finite sum gives

$$\frac{1}{|H|} \sum_{g \in H} f(g) = \sum_{j \in J} \frac{|C_j|}{|H|} f(u_j).$$

Every coefficient $w_j = |\mathcal{C}_j|/|H|$ is nonzero. In a character calculation, the required constancy and vanishing concern the product of the two functions in the scalar product.

Implementation.

`ManuscriptIBAW.Library.FiniteClassPairing` proves `FiniteClassPairing.sum_localize` for a finite set with an explicit class map and distinct selected class labels. It uses constancy on the selected fibres and vanishing elsewhere. The theorem `FiniteClassPairing.weight_ne_zero` proves nonvanishing of the normalised fibre sizes in characteristic zero, using a representative in each fibre. These results do not require a character theory. Let K be a field, let J be finite, and let the vectors $v_i \in K^J$, indexed by a set I , span K^J . Suppose that $w_j \neq 0$ for every j and that $z_r \in K^J$ is nonzero for each $r \in J$. Let $a_{i,r} \in K$ and $\varepsilon_i \in K$ satisfy

$$\sum_{j \in J} w_j v_i(j) z_r(j) = \varepsilon_i a_{i,r} \quad (i \in I, r \in J).$$

Then no column of $(a_{i,r})$ is zero. Indeed, if the column at r were zero, the linear functional $v \mapsto \sum_j w_j v(j) z_r(j)$ would vanish on a spanning family. It would therefore vanish on every coordinate vector, giving $w_j z_r(j) = 0$ for all j . This contradicts $z_r \neq 0$.

Suppose now that K has characteristic zero, every $a_{i,r}$ is a natural number viewed in K , and each row has sum one. Every row then has exactly one entry equal to one and all others zero. Since every column is nonzero, choose $s(r) \in I$ with $a_{s(r),r} = 1$. The map $s : J \rightarrow I$ is injective and satisfies

$$a_{s(r),j} = \begin{cases} 1 & j = r, \\ 0 & j \neq r. \end{cases}$$

Injectivity follows by comparing the unique nonzero entries in two selected rows.

Implementation.

In `ManuscriptIBAW.Library.FiniteCharacterSelection`, `FiniteCharacterSelection.coverage_of_span` proves that every column is nonzero. The theorem `FiniteCharacterSelection.exists_delta_injection_of_span` constructs the injective selection assuming that the entries are nonnegative integers and each row sums to one. These are finite linear algebra and arithmetic results over the stated fields. `span_eq_top_of_diagonal_pairings` and `span_eq_top_of_trace_expansions` derive the spanning condition from diagonal pairing identities with nonzero diagonal entries and from explicit finite trace expansions, respectively. `linearIndependent_of_spanning_independent_family` proves that a finite family is linearly independent if its span contains a linearly independent family of the same size. In the character application, the identification of these families, expansions and scalar products with the geometric and representation theoretic results remains a separate set of assumptions.

The row sum condition can also be proved by counting. Let I and J be finite sets with $I \neq \emptyset$, and let U be a set. Let $f : J \rightarrow U$, $u \in U$, and $a : I \times J \rightarrow \mathbb{N}$. Suppose that the sum in column j is one when $f(j) = u$ and zero otherwise, and that $|I| = |f^{-1}(u)|$. If every two rows are related by a permutation of J , then every row has sum one. Indeed, all row sums have a common value r . Summing the entries by rows and by columns gives

$$|I|r = |f^{-1}(u)| = |I|,$$

and $I \neq \emptyset$ implies $r = 1$.

Implementation.

`ManuscriptIBAW.TypeC.PrincipalGGGR.row_sum_one` proves this finite arithmetic statement. It takes the column sums, the equality of cardinalities and the row permutations as assumptions.

Chapter 11

Group actions, characters and block parameters

Equivariant maps and character identities often determine an action without requiring an explicit choice of representatives. We study this principle for permutation modules, block stabilisers and ordinary characters over different fields. We also describe the quotient associated with a similitude multiplier and the relation between block idempotents and rational series.

11.1 Fixed bases and permutation modules

Let A act linearly on a module V over a nonzero ring R . If A fixes every vector of an R -basis, it acts trivially on V : a linear map is determined by its values on a basis. If another basis is indexed by an A -set Y and the action permutes its basis vectors in accordance with the action on Y , the linear independence of that basis implies that A fixes Y pointwise.

Proposition 11.1. *Let X and Y be finite A -sets. Suppose that $d: \mathbb{Z}X \rightarrow \mathbb{Z}Y$ is an A -equivariant isomorphism of free abelian groups. If A fixes X pointwise, it fixes Y pointwise, and there is an A -equivariant bijection $X \rightarrow Y$.*

Proof. The images under d of the basis vectors indexed by X form a pointwise fixed basis of $\mathbb{Z}Y$. The preceding argument shows that A acts trivially on $\mathbb{Z}Y$, and hence on Y . Equality of the ranks gives $|X| = |Y|$. Any bijection between the two finite sets is then equivariant. $\square$

Let X be an A -stable integral basic set of a block b , and suppose that the restriction of the decomposition homomorphism gives an A -equivariant isomorphism $\mathbb{Z}X \rightarrow \mathbb{Z} \text{IBr}(b)$. If A fixes every character in X , the proposition shows that it fixes every irreducible Brauer character of b and gives an A -equivariant bijection $X \rightarrow \text{IBr}(b)$.

Implementation.

`ManuscriptIBAW.Library.FixedBasis` specialises Mathlib's basis extensionality, injectivity and equivalence of basis index sets to the stated actions. `ManuscriptIBAW.Library.FixedBasicSet` applies this argument to Mathlib's integral permutation modules. The block application is `ManuscriptIBAW.TypeC.EvenBasicSet`.

11.2 Equivariant bijections for involutions

Proposition 11.2. *Let A be a group acting on finite sets X and Y . Suppose that A is generated by a set T and an element δ , that every element of T fixes both sets pointwise, and that δ induces an involution on each set. There is an A -equivariant bijection $X \rightarrow Y$ if*

$$|X| = |Y| \quad \text{and} \quad |X^{(\delta)}| = |Y^{(\delta)}|.$$

Proof. Each orbit under δ has size one or two. The two equalities give equal numbers of orbits of each size. Match the fixed points and the orbits of size two to obtain a bijection commuting with δ . It commutes with T because T acts trivially. The elements of A with which the bijection commutes form a subgroup, so the generating hypothesis gives equivariance under A . $\square$

The hypothesis concerns the permutations induced by δ . It does not require that δ itself have order two in A . This distinction allows the proposition to be used when δ^2 is an inner automorphism and the objects under consideration are conjugacy classes.

Implementation.

`ManuscriptIBAW.Library.EquivariantBijections` proves the subgroup and generating set arguments and uses `Formalisation.C2Cancellation` for the two involutions. That result specialises Mathlib's conjugacy classification of finite permutations by cycle type and its fixed point count. The local statements express the conclusion as an equivariant bijection for the given actions.

11.3 Independent signs on a direct sum

Let $V = \bigoplus_{c \in I} V_c$ be a finite dimensional space over a field of characteristic different from two. For each c , let ε_c act as -1 on V_c and as 1 on the other summands. A linear map commuting with every ε_c preserves every V_c . Indeed, in these coordinates, commutation forces an entry between distinct summands to equal its negative. That entry is zero because two is invertible.

If the decomposition is orthogonal for a nondegenerate alternating form and the map is symplectic, its restriction to each summand is symplectic. The form equation restricts to each diagonal block. Consequently the map belongs to the product of the symplectic groups of the summands.

Suppose that $R = \prod_c R_c$ acts independently on these summands, that $-1 \in R_c$, and that the centraliser of R_c in its symplectic group is contained in R_c . Every element centralising R commutes with the independent signs, hence preserves each summand. Its restriction there centralises R_c and therefore belongs to R_c . Thus $C_{\text{Sp}(V)}(R) \leq R$. The index c distinguishes different summands, including copies on which the same group acts.

Implementation.

`ManuscriptIBAW.TypeC.independentSignSeparation` in `ManuscriptIBAW.TypeC.PrincipalProducts` proves the matrix argument for the specified symplectic coordinates over a finite field of odd order. Mathlib supplies matrix reindexing and the block diagonal multiplication and transpose identities. The proof combines them with the specified embedding and Gram matrix equations to separate the summands.

`ManuscriptIBAW.TypeC.PrincipalProductData.signs` is a proved consequence of those equations. The local centraliser assertion remains a source assumption in the application.

11.4 Stabilisers under an equivariant bijection

Let $A = M \rtimes E$ act on sets X and Y , and let $\beta : X \rightarrow Y$ be an A -equivariant bijection. Then $A_x = A_{\beta(x)}$, since equivariance and injectivity give $\beta(ax) = \beta(x)$ if and only if $ax = x$. In particular, if

$$m(e\beta(x)) = \beta(x) \iff m\beta(x) = \beta(x) \text{ and } e\beta(x) = \beta(x),$$

the same equivalence holds with x in place of $\beta(x)$. This transfers the factorisation of a stabiliser into its M and E parts.

Chapter 9 constructs a global bijection from bijections over representatives of block orbits that are equivariant under their stabilisers. It preserves the block assignments. For character sets of a finite group G , suppose that A acts through the usual automorphism action and maps onto $\text{Aut}(G)$. Equivariance under A then gives equivariance under every automorphism: choose a preimage and apply the same identity.

Implementation.

`ManuscriptIBAW.TypeC.OddGlobalFactorization` constructs `globalBijection` from the block basic sets, the stated quotient actions and the Conlon and Burnside hypotheses. In that namespace, `GlobalBijection.automorphism_equivariant` proves full automorphism equivariance, and `GlobalBijection.allBrauerFactorization` transfers ordinary stabiliser factorisation through that same bijection. The construction uses the basic set hypotheses first, then transfers the factorisation to the Brauer characters.

11.5 Linear characters and block stabilisers

Under the coefficient and tensor compatibility hypotheses of Section 7.6, let G be finite, let ℓ be prime and let $N \triangleleft G$. Let b be an ℓ -block of G , and let C be a subgroup of the group of ordinary linear characters of G/N of order prime to ℓ . Inflate these characters to G and let them act by tensoring. Each character in C_b is trivial on $Z(G)$, so it factors through $G/NZ(G)$. In particular,

$$|C_b| \leq |G : NZ(G)|.$$

The same assertion holds for the corresponding linear Brauer characters. The proof uses the common central character of the irreducible Brauer characters of b . A character stabilising b is trivial on the ℓ' -part of $Z(G)$ by this central character, and on its ℓ -part because its order is prime to ℓ .

Now let a finite cyclic group E act on C , and suppose that $C \rtimes E$ acts on the blocks in a way whose restriction to C is the tensor action just described. For $J = (C \rtimes E)_b$, let D be the kernel of the projection $J \rightarrow E$. Sending an element of D to its C coordinate gives an injection $D \rightarrow C_b$. Thus $|D| \leq |G : NZ(G)|$. If this index is at most two, D is a normal 2-subgroup of J and J/D is cyclic.

Let P/D be the Sylow 2-subgroup of J/D . Then $P \triangleleft J$ is a 2-group and J/P is cyclic of odd order. For every subgroup $U \leq J$, the subgroup $U \cap P$ is a normal 2-subgroup and $U/(U \cap P)$ embeds in J/P . Consequently $U/O_2(U)$ is cyclic of odd order. Thus every subgroup of J is 2-hypoelementary, as required in the application of Conlon's theorem.

Implementation.

The central character argument is in `ModularRep.LinearBlockStabilizer`.

`ManuscriptIBAW.Library.TensorStabilizer` identifies the kernel of the projection to E with its character coordinate. It combines the resulting bound with the hypoelementary group calculation in `ModularRep`. Compatibility of the block action with tensoring simple modules remains an explicit hypothesis.

11.6 The multiplier quotient

Proposition 11.3. *Let $\mu: G \rightarrow U$ be a surjective homomorphism to an abelian group. It induces an isomorphism*

$$G/(\ker \mu Z(G)) \cong U/\mu(Z(G)).$$

If $U = \mathbb{F}_q^\times$ and $\mu(Z(G)) = (\mathbb{F}_q^\times)^2$, this quotient has order at most two, and has order two when q is odd.

Proof. The composite $G \rightarrow U \rightarrow U/\mu(Z(G))$ is surjective with kernel $\ker \mu Z(G)$, so the first assertion is the first isomorphism theorem. In $\mathbb{F}_q^\times$, the kernel of the squaring map is the group of roots of $x^2 = 1$. It has at most two elements and has exactly two in odd characteristic. The index of the image of an endomorphism of a finite group equals the order of its kernel. $\square$

For the conformal symplectic group of positive rank, the multiplier is specified by $\langle gv, gw \rangle = \mu(g)\langle v, w \rangle$. Its kernel is the symplectic subgroup. Surjectivity and the image of the centre are the structural hypotheses in this application. The proposition determines the quotient index from these hypotheses.

Implementation.

`ManuscriptIBAW.Library.MultiplierQuotient` obtains the isomorphism by identifying the kernel of the composite and applying Mathlib's first isomorphism theorem. The finite field calculation combines Mathlib's index formula for an endomorphism with its bounds and exact counts for roots of unity. `ManuscriptIBAW.TypeC.ConformalMultiplier` applies them to the matrix groups with the specified symplectic form.

11.7 Ordinary characters over a common cyclotomic field

Let G be finite, let N be a positive multiple of $|G|$, and set $L = \mathbb{Q}(\zeta_N)$. Fix embeddings $\iota_{\mathbb{C}}: L \rightarrow \mathbb{C}$ and $\iota_K: L \rightarrow K$, where K has characteristic zero. For character independence and the criterion for realisability, see, for example, Serre, Section 12.1, Propositions 32–33. The cyclotomic realisation theorem appears there as Theorem 24 (Brauer), with its corollary, in Section 12.3 [8].

Assume that a realisation statement supplies, for each complex irreducible character χ , characters $\chi_L \in \text{Irr}_L(G)$ and $\chi_K \in \text{Irr}_K(G)$. If the chosen basic set associates a simple $K[G]$ -module V_χ with χ , the character of its representation ρ_χ must equal χ_K . Thus the required identities are

$$\iota_{\mathbb{C}}(\chi_L(g)) = \chi(g), \quad \iota_K(\chi_L(g)) = \chi_K(g) = \text{tr}_K(\rho_\chi(g)) \quad (g \in G).$$

These identities relate the two realisations to the selected labels. They can also be obtained by constructing those labels from the realisation statement.

Suppose that a group of automorphisms permutes a chosen set of complex characters. Injectivity of $\iota_{\mathbb{C}}$ transports the character identities to L , and applying ι_K gives the same identities over K . Equality of the traces of the corresponding irreducible representations gives their isomorphism over K , and hence equality of their classes in the exact Grothendieck group. Thus the action on the ordinary labels used by the decomposition homomorphism follows from the action on the complex characters and the stated realisations.

The argument uses only the two embeddings of L , so it requires neither an embedding of $\mathbb{C}$ into K nor algebraic closure of K . Existence of the realisations is an assumption. Their compatibility with automorphisms follows from these trace identities.

Implementation.

The common field is Mathlib's cyclotomic field. Trace separation uses Mathlib's character averaging formula for the dimension of the intertwiner space and its theorem that a nonzero intertwiner between irreducibles is bijective. ModularRep combines these results over the stated field K . In `ManuscriptIBAW.Characters.CyclotomicOrdinaryLabels`, `OrdinaryLabelRealisation.ordinary_value` is the trace identity. For labels obtained from source realisations, `SerreRealisationSource.labelRealisation` proves it.

11.8 Primitive blocks and rational series

Let $(b_B)_{B \in \mathcal{B}}$ be a finite complete family of nonzero orthogonal central idempotents of a ring. Assign to each B a parameter t_B in a finite group H^* . For $t \in H^*$, define

$$e_t = \sum_{\substack{B \in \mathcal{B} \\ t_B \sim t}} b_B,$$

where $\sim$ denotes conjugacy in H^* . Orthogonality gives

$$b_B e_t = \begin{cases} b_B, & t_B \sim t, \\ 0, & t_B \not\sim t. \end{cases}$$

Since $b_B \neq 0$, we have $b_B e_t = b_B$ if and only if t_B is conjugate to t .

In Jordan reduction the idempotents are the blocks of the specified group algebra. Each t_B is required to be semisimple and of order prime to ℓ . Let s index a geometric series, and write $t_s \in H^*$ for its assigned element of the finite dual group. The geometric series idempotent at s is identified with this same sum e_{t_s} . Completeness of the chosen semisimple parameters then places every block in one of the series.

The label used to test strict quasi-isolation is also identified with the image of t_B under a specified injective homomorphism from H^* to the algebraic dual group. The intended map includes rational points in the algebraic group. Compatibility of series with automorphisms is a separate identity for the induced map of group algebras and the dual action. These identifications specify the blocks, labels and action to which Jordan reduction applies. Their interpretation as rational series in algebraic groups remains an assumption.

Implementation.

`ManuscriptIBAW.Jordan.SeriesBinding` defines `rationalSeriesIdempotent` and the identification record `SeriesBinding` using the block family in `ModularRep`. Mathlib supplies the group algebra and finite sum operations. The local calculation applies orthogonality to this specified sum, while its identification with the geometric series remains a hypothesis. The multiplication formula is `block_mul_rationalSeriesIdempotent`. The membership equivalence and coverage of the blocks are `SeriesBinding.block_belongs_iff` and `SeriesBinding.every_block_belongs`, all in `ManuscriptIBAW.Jordan`.

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